

Polymer-Based Solid-State Electrolytes for High-Energy-Density Lithium-Ion Batteries – Review

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Portable electronic devices and electric vehicles have become indispensable in daily life and caused an increasing demand for high-performance lithium-ion batteries (LIBs) with high-energy-density. This work compares the intrinsic characteristics and Li^+ conduction mechanisms of various electrolytes, aiming at emphasizing their suitability for high-energy-density LIBs. Among all electrolytes, polymer-based solid-state electrolytes (SSEs) are the most promising candidates, as they demonstrate the most comprehensive properties. The advantages and disadvantages of commonly used polymer matrix materials of SSEs are discussed, along with typical approaches to address their limitations. As significant issues for high-energy-density and cycle stability, the development related to the cathode/electrolyte interfacial contact and wetting, interfacial electrochemical compatibility, and interfacial Li^+ conduction in LIBs employing polymer-based SSEs, as well as the anode/electrolyte interfacial chemical stability and lithium dendrite suppression are comprehensively reviewed and analyzed. Finally, perspectives on future research directions for developing high-energy-density LIBs are highlighted building upon the existing literature.

1. Introduction

Lithium-ion batteries (LIBs) have many advantages including high-operating voltage, long-cycle life, and high-energy-density, etc.,^[1] and therefore they have been widely used in portable electronic devices, electric vehicles, energy storage systems, and other special domains in recent years, as shown in Figure 1.^[2–4] Since the Paris Agreement has been put into effect in November 2016, the market demand for LIBs continued to rise. As expected, the global shipments of LIBs will reach 439.32 GWh in 2025.^[5]

At present, almost all LIBs use liquid electrolytes (LEs) as the Li^+ conductors. Although they have high-ionic conductivity and good contact with the electrode, their thermal stability and flame point are low. The spontaneous formation of lithium dendrites during the Li^+ deposition and stripping process in LEs may cause short circuits of the batteries, and even lead to critical safety issues like fire and explosion

accidents if they are improperly used.^[6,7] Consequently, enhancing the safety performance and increasing the energy-density of LIBs, especially for portable electronic devices and electric vehicles, has become an urgent priority.

To address these challenges, safe solid-state electrolytes (SSEs) have been proposed and developed. SSEs offer good mechanical strength and wide electrochemical stability windows, and solid-state lithium-ion batteries (SSLIBs) require simplified packaging.^[8] Furthermore, the thinness of SSEs allows high-energy-density for SSLIBs.^[9] On top of this, it is anticipated that by augmenting the mass-loading of the cathodes in SSLIBs, even greater energy-density can be achieved.

However, maintaining a close and intimate cathode/electrolyte interfacial contact remains a challenge in SSLIBs. Unlike the conventional LEs, most SSEs struggle to wet and penetrate the porous cathodes, resulting in limited Li^+ conduction within the cathodes and across the cathode/electrolyte interface. This limitation hampers the utilization of high-mass-loading thick cathodes. Among SSEs, polymer-based SSEs have shown the most promising interfacial contact with electrodes, owing to their soft nature and desirable flexibility.^[10] In situ polymerization of polymers and the addition of certain liquid components offer potential solutions to enhance the wetting of the thick cathodes and

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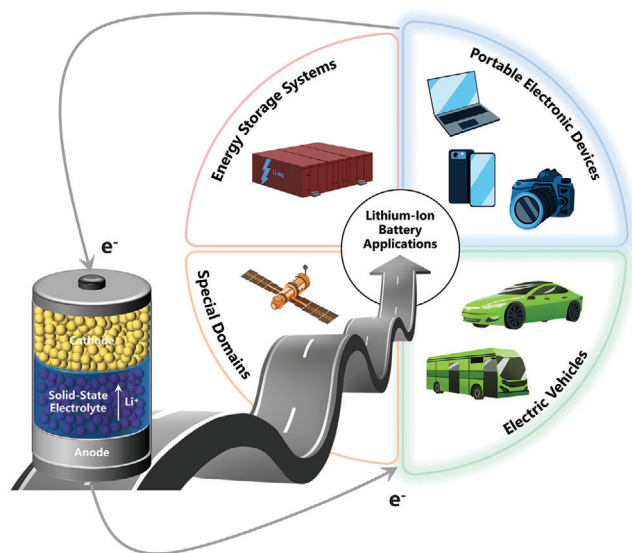


Figure 1. Typical applications of LIBs.

enable the achievement of higher energy-density of SSLIBs. Nevertheless, the cathode/electrolyte interfacial electrochemical stability and the presence of space-charge layers remain significant issues that strongly influence the energy-density and cycle performance of SSLIBs with polymer-based SSEs. In addition, the lithium metal anode/electrolyte interface also has the problem of chemical stability and lithium dendrite suppression. The development of high-energy-density SSLIBs still has formidable challenges.

In this work, we conduct a comprehensive review of recent research on polymer-based SSEs for high-energy-density SSLIBs. We initially summarize and analyze the intrinsic characteristics and Li^+ conduction mechanisms of various electrolytes. Afterward, we discuss the advantages and disadvantages of the extensively studied polymer materials used in polymer-based SSEs, along with the typical modification methods employed to address their limitations. Subsequently, we analyze the existing issues of electrode/polymer-based SSE interfaces and their remedies. Finally, we provide insights into potential avenues for further improving the energy-density of SSLIBs with polymer-based SSEs.

2. Intrinsic Characteristics and Li^+ Conduction Mechanisms of Various Electrolytes

LEs and SSEs represent the two fundamental categories of Li^+ conductors. SSEs can be further divided into two specific types: solid polymer electrolytes (SPEs) and inorganic ceramic electrolytes (ICEs). Each type of electrolyte possesses its advantages/disadvantages with its distinct Li^+ conduction mechanism.^[11–13] To enhance the overall properties, such as ionic conductivity, electrochemical/thermal stability, and mechanical strength, LEs, SPEs, and ICEs have been combined to form new electrolytes consisting of two or even three components. **Figure 2** provides a comprehensive overview of the complex electrolytes, which will be elaborated upon in the following sections.

Complex Li^+ conduction mechanisms exist in the electrolytes containing multiple components, as each component will contribute a distinct Li^+ conduction pathway. Interactions between different components further complicate the Li^+ conduction mechanisms, resulting in both synergistic and competitive effects. A comprehensive understanding of Li^+ conduction mechanisms in various electrolytes facilitates the design of electrolytes with high-ionic conductivity at a theoretical level. Additionally, it enables the optimization of the cathode/electrolyte interface for high-energy-density LIBs. As known, both the intrinsic ionic conductivity of the electrolyte itself and the cathode/electrolyte interfacial conductivity play crucial roles in high-performance LIBs with the aim of outstanding energy-density.

Ionic conductivity (σ) can be expressed as the reciprocal of ionic resistivity (ρ) which is a function of thickness (l), cross-sectional area (A) and the bulk resistance (R_0) measured between two blocking electrodes as Equation (1) shows:

$$\sigma = \frac{1}{\rho} = \frac{l}{R_0 A} \quad (1)$$

Most electrolytes are dual-ion (Li^+ and anions) conductors. Therefore, it is necessary to define the Li^+ transference number (t^+) as the proportion of Li^+ mobility (μ^+) and the total mobility [including anion mobility (μ^-)] as Equation (2) shows:

$$t^+ = \frac{\mu^+}{\mu^+ + \mu^-} \quad (2)$$

In addition, ionic conductivity can also be calculated by the sum of the product of charge carrier concentration (n_i), number of charges (q_i) and mobility (μ_i) as Equation (3) shows:

$$\sigma = \sum n_i q_i \mu_i \quad (3)$$

2.1. Single-Component Electrolytes

2.1.1. Liquid Electrolytes

LEs possess distinct advantages and disadvantages. They exhibit the highest ionic conductivity among all electrolytes (ranging from 10^{-2} to $10^{-3} \text{ S cm}^{-1}$ at room temperature), and their liquid nature enables the effective wetting of electrodes, resulting in excellent electrode/electrolyte interfacial contact. However, LEs often suffer from low-thermal/electrochemical stability. Moreover, LEs lack the ability to suppress lithium dendrite due to their limited mechanical strength. The requirement for separators in LE-based LIBs adds complexity to their overall structure compared to SSLIBs. All these shortcomings make the safety of LEs a significant concern and impede the achievement of high-performance LE-based LIBs.

As shown in **Figure 3a**, the limited electrochemical stability window of LEs is evident in the narrow gap between their reduction potential at negative potentials and solvent oxidation potential at positive potentials.^[14] However, to maximize the energy-density of LIBs, LEs are required to serve the high-voltage cathodes, and the operation voltage may be beyond the electrochemical stability window of the LEs. To enhance the cathode/LE

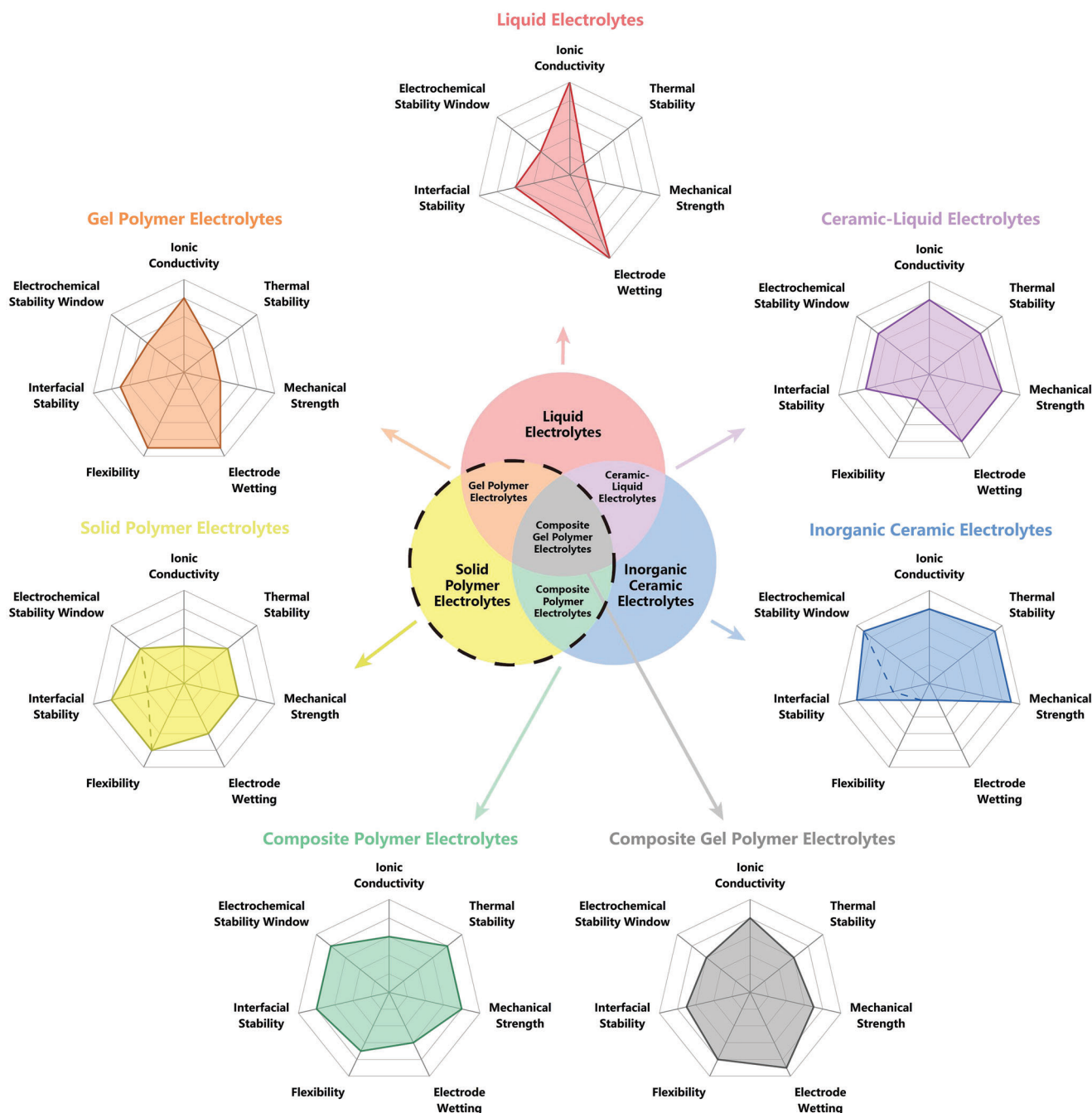


Figure 2. Distinguished properties of various electrolytes. Note: The composite polymer electrolytes (CPEs) and composite gel polymer electrolytes (CGPEs) may contain either ionic conductive ICEs or non-ionic conductive ceramic fillers. The gel polymer electrolytes (GPEs) and CGPEs may include LEs, ionic liquids (ILs), or other functional solvents as plasticizers. The dashed line in SPEs and ICEs represents several candidates that are not stable toward lithium metal anodes, and the instability remains when they are used in other types of SSEs. The dashed circle of polymer-based SSEs indicates they are most suitable for high-energy-density SSLIBs, which will be explained in the following section.

interfacial stability, a stable cathode/electrolyte interphase (CEI) is usually designed to form at the interface, which involves the decomposition and phase transformation of both the cathode and electrolyte.^[15] On the other hand, the stable solid-electrolyte interphase (SEI) at the anode/electrolyte interface can be formed through the reductive decomposition of the solvation sheath,

where the composition of the solvation species influences the stability of the SEI.^[15–17]

During charging, Li^+ is released from the cathode and will pass through the CEI and LE toward the anode (Figure 3b). Within the LEs, the Li^+ is surrounded by several solvent molecules, forming a solvation sheath known as the solvation of Li^+ .

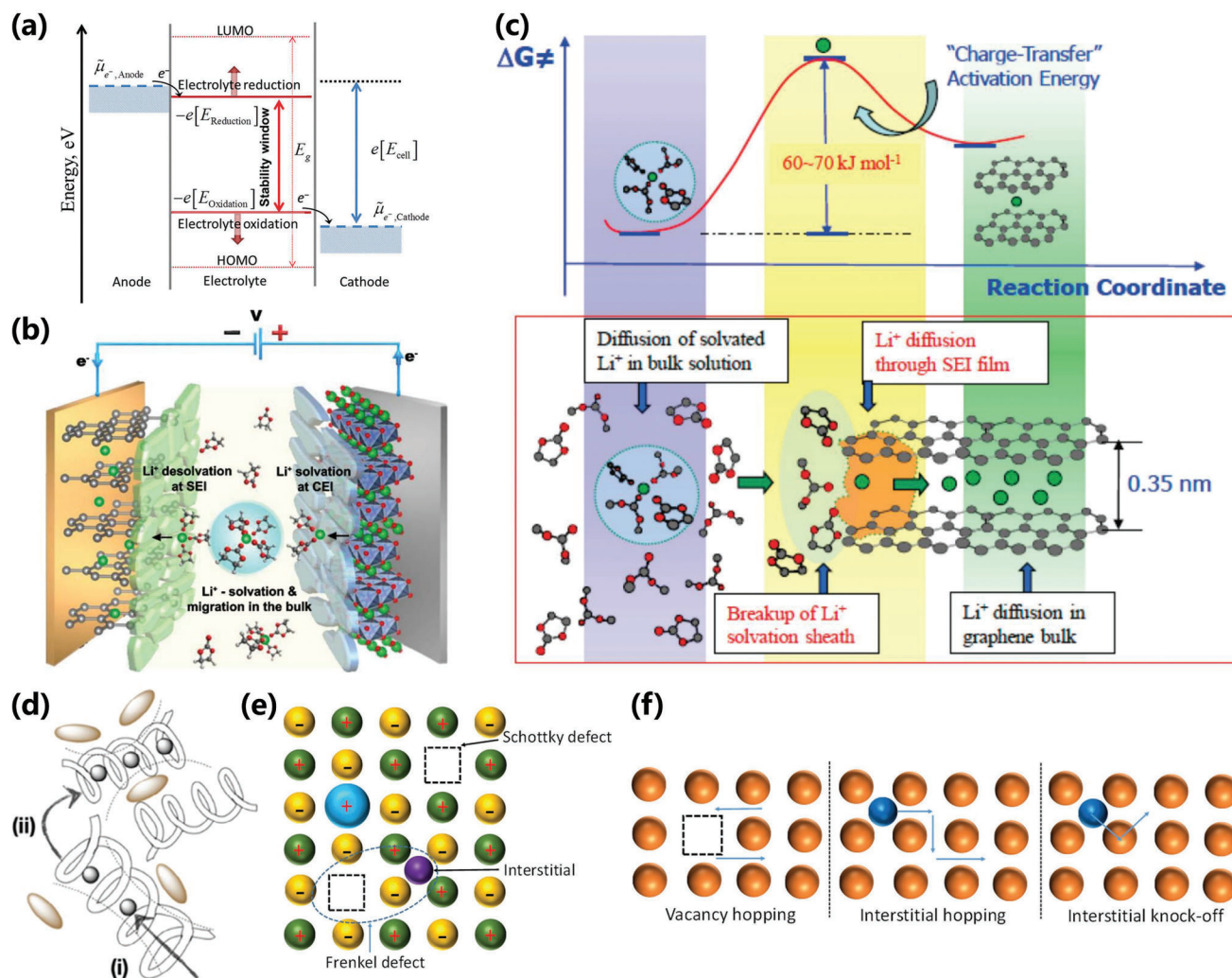


Figure 3. a) Negative/positive potential limitations for LEOs. Reproduced with permission.^[14] Copyright 2018, Royal Society of Chemistry. b) Li⁺ conduction in LIBs using LEOs. Reproduced with permission.^[15] Copyright 2022, American Association for the Advancement of Science. c) The rate-determining step of the Li⁺ conduction at the electrode/LE interfaces. Reproduced with permission.^[19] Copyright 2010, American Chemical Society. d) Li⁺ conduction mechanism in SPEs. Reproduced with permission.^[28] Copyright 2020, John Wiley and Sons. e) Typical defects in ICs. Reproduced with permission.^[46] Copyright 2018, John Wiley and Sons. f) Different types of Li⁺ conduction in ICs. Reproduced with permission.^[46] Copyright 2018, John Wiley and Sons.

This core (Li⁺)-shell (solvent molecules) structure migrates together. Upon reaching the anode, the Li⁺ is de-solvated from the solvation sheath and adsorbed onto the surface of the SEI/anode. Subsequently, the adsorbed Li⁺ undergoes migration through the SEI and participates in the charge transfer reactions at the anode. This process is reversed during discharging. It should be noted that the de-solvation of Li⁺ needs to overcome a significant energy barrier (Figure 3c), thus influencing the overall rate of interfacial Li⁺ conduction.^[18,19] In addition to the solvent molecules, the anions from some lithium salts may also participate in the solvation of Li⁺,^[20] and compared to the solvent molecule sheath, the anion sheath displays a lower de-solvation energy barrier and has the ability to assist the formation of stable inorganic-rich SEIs and CEIs. Therefore, the dominance of anions in the solvation sheaths is preferred.^[16,17]

2.1.2. Solid Polymer Electrolytes

SPEs exhibit remarkable processability and flexibility.^[21,22] Although they may not wet the electrodes as effectively as LEOs, their flexibility allows for favorable contact with the electrodes. Certain polymers can be in situ polymerized, leading to even better electrode/electrolyte interfacial properties. However, their soft nature also makes them unsatisfactory for suppressing lithium dendrites.^[21] Also, their ionic conductivity (ranging from 10^{−8} to 10^{−6} S cm^{−1} at room temperature) is far below the practical application requirements.^[23] Even though SPEs have a wider electrochemical stability window than LEOs, they still face challenges in matching the high-voltage cathodes. Furthermore, some polymers, like polyacrylonitrile (PAN) and poly(vinylidene fluoride) (PVDF), are unstable with the lithium metal anodes due to their specific functional groups.

Most SPEs are based on crystalline polymers, and the conduction of Li^+ mainly occurs within the non-crystalline regions of the polymers.^[24] Take an example of the poly(ethylene oxide) (PEO), the dissociated Li^+ will coordinate with oxygen atoms from the ethylene oxide groups on the PEO main chain and migrate in conjunction with the segmental motion of the chains under an electric field.^[25] Consequently, the segment motion of the polymer chain directly affects the Li^+ conduction. Within the SPEs, Li^+ hop within or between chains, accompanied by the breakage and formation of coordination bonds.^[26–28] This simultaneous intra-chain [Figure 3d(i)] and inter-chain [Figure 3d(ii)] hopping of Li^+ , coupled with the segmental motion of the polymer chains, ultimately enables the polymer to conduct Li^+ .^[28–30]

In SPEs, the activation energy for ion migration can be derived from the relationship between the ionic conductivity and temperature, which generally follows two empirical models: Arrhenius model or Vogel-Tamman-Fulcher (VTF) model. The Arrhenius equation is applicable to the situation when the Li^+ motion is independent of the molecular motion of the polymer host. As Equation (4) shows, C is the pre-exponential factor, which is related to the number of carriers, k_B is the Boltzmann constant and E_a is the activation energy, which means the energy barrier for Li^+ migrating from its original site to the adjacent sites and can be obtained by linear fitting of $\log \sigma$ and $\frac{1}{T}$.

$$\sigma = C e^{-\frac{E_a}{k_B T}} \quad (4)$$

The VTF equation is, on the other hand, applicable to the situation when Li^+ transport is strongly associated with the segmental relaxation in polymers. As Equation (5) shows, σ_0 is the pre-exponential factor, B is related to the activation energy ($B = \frac{E_a}{k}$) and $T_0 = T_g - 50 \text{ K}$. In this situation, the segmental motion of polymers will be accelerated at elevated temperatures which offers additional inter/intra-chain Li^+ hopping sites. The VTF behavior will be often observed in the CPEs and GCPEs. However, the linear Arrhenius phenomenon with lower E_a can still be observed in some SPEs at higher temperatures. This is because, with the increase in temperature, the polymer segmental motion brings the hopping sites closer, and the ions can therefore migrate from one occupied site to another empty site with less energy. In other words, the polymer segmental motion is decoupled from the ionic motion.^[30]

$$\sigma = \sigma_0 T^{-\frac{1}{2}} e^{-\frac{B}{T-T_0}} \quad (5)$$

2.1.3. Inorganic Ceramic Electrolytes

Typical ICEs can be grouped into oxides, sulfides, argyrodites, halides, etc. Among which oxide ICEs have been discovered and researched for decades, including NASICON-type $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ (LAGP), LISICON-type $\text{Li}_{14}\text{Zn}(\text{GeO}_4)_4$, garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) and perovskite-type $\text{Li}_{0.5}\text{La}_{0.5}\text{TiO}_3$ (LLTO), etc.^[4] They have also been widely used as ceramic fillers in CPEs and GCPEs. Oxide ICEs have an ionic conductivity comparable to LEs. They also offer high-electrochemical/thermal stability, and their high-mechanical strength can effectively inhibit lithium dendrite

growth.^[31] However, a notable drawback of the oxide ICEs is their inferior flexibility and the point-to-point contact against electrodes.^[32] Notably, degradation may happen to the oxide ICEs containing titanium and germanium elements [like $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$ (LATP), LAGP, and LLTO] when in contact with lithium metal anodes due to the reduction of titanium and germanium.^[33–35]

Sulfide ICEs exhibit a distinct structural resemblance to the LISICON-type oxide ICE ($\gamma\text{-Li}_3\text{PO}_4$), wherein S^{2-} replaces O^{2-} . Due to the larger radius and enhanced polarizability of S^{2-} over O^{2-} , the interaction between S^{2-} and Li^+ is comparatively weaker. Consequently, the sulfide ICEs display a higher ionic conductivity.^[4,36–41] Furthermore, the soft nature of the sulfide ICEs also allows better contact with electrodes.^[41] Argyrodite ICEs $\text{Li}_6\text{PS}_5\text{X}$ ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) have emerged as a crucial family of sulfide ICEs, showing even higher ionic conductivity (exceeding $10^{-3} \text{ S cm}^{-1}$ at room temperature).^[37–41] This is because the disorder in $\text{S}^{2-}/\text{Cl}^-$ or $\text{S}^{2-}/\text{Br}^-$ can promote the mobility of Li^+ . However, it should be noted that the air stability of the sulfide and argyrodite ICEs is poor, limiting their practical applications to a large extent.^[39–41]

Unlike sulfide ICEs, halide ICEs [Li_3MX_6 (M^{3+} , $\text{X} = \text{Cl}, \text{Br}, \text{I}$)] can remain stable in both dry air and humid atmospheres. Furthermore, the electrochemical stability window of such electrolytes can reach 6.0 V (vs Li/Li^+). However, the ionic conductivity of the previously reported halide ICEs is not satisfied. Recently, halide ICEs with improved ionic conductivity ($\approx 10^{-3} \text{ S cm}^{-1}$) have been developed.^[42–44]

Li^+ conduction in ICEs heavily relies on the point defects present in the crystal structure, including Schottky defects (anion vacancies accompanied by a cation vacancy) and Frenkel defects (vacancies accompanied by an interstitial ion) as shown in Figure 3e.^[26,45–46] Within the ICEs, Li^+ can hop between the vacancies created by the Schottky defects, and once they move to a new vacancy, the original vacancy becomes available for other Li^+ . Through Frenkel defects, the interstitial Li^+ can diffuse by continuously displacing the Li^+ in the nearby interstitial sites (known as interstitial hopping) or by occupying the lattice site along with pushing the origin lattice Li^+ into the adjacent interstitial site (known as the interstitial knock-off).^[45–47] Therefore, the Li^+ conduction mechanism in ICEs can be summarized as vacancy hopping, interstitial hopping, and interstitial knock-off (Figure 3f).^[26,46] and the ionic conductivity of ICEs is determined by the carrier concentration, diffusion types/pathways, all of which are closely related to the concentration and distribution of the defects within the crystals of ICEs. The activation energy for ion migration within ICEs can be derived from the Arrhenius equation [Equation (4)].

Within the polycrystalline ICEs, Li^+ will migrate via grains as well as boundaries, among which the grain boundary resistance usually surpasses the intragrain resistance, making boundary conduction the primary limiting factor. This increased resistance leads to an enlarged energy barrier for the Li^+ migration, thereby reducing the overall ionic conductivity of ICEs. The ionic conductivity of the oxide and halide ICEs is notably affected by the presence of high boundary resistance. In contrast, sulfide ICEs exhibit higher ionic conductivity due to their lower boundary resistance. This favorable characteristic can be attributed to the synthesis conditions and the soft nature of sulfides.

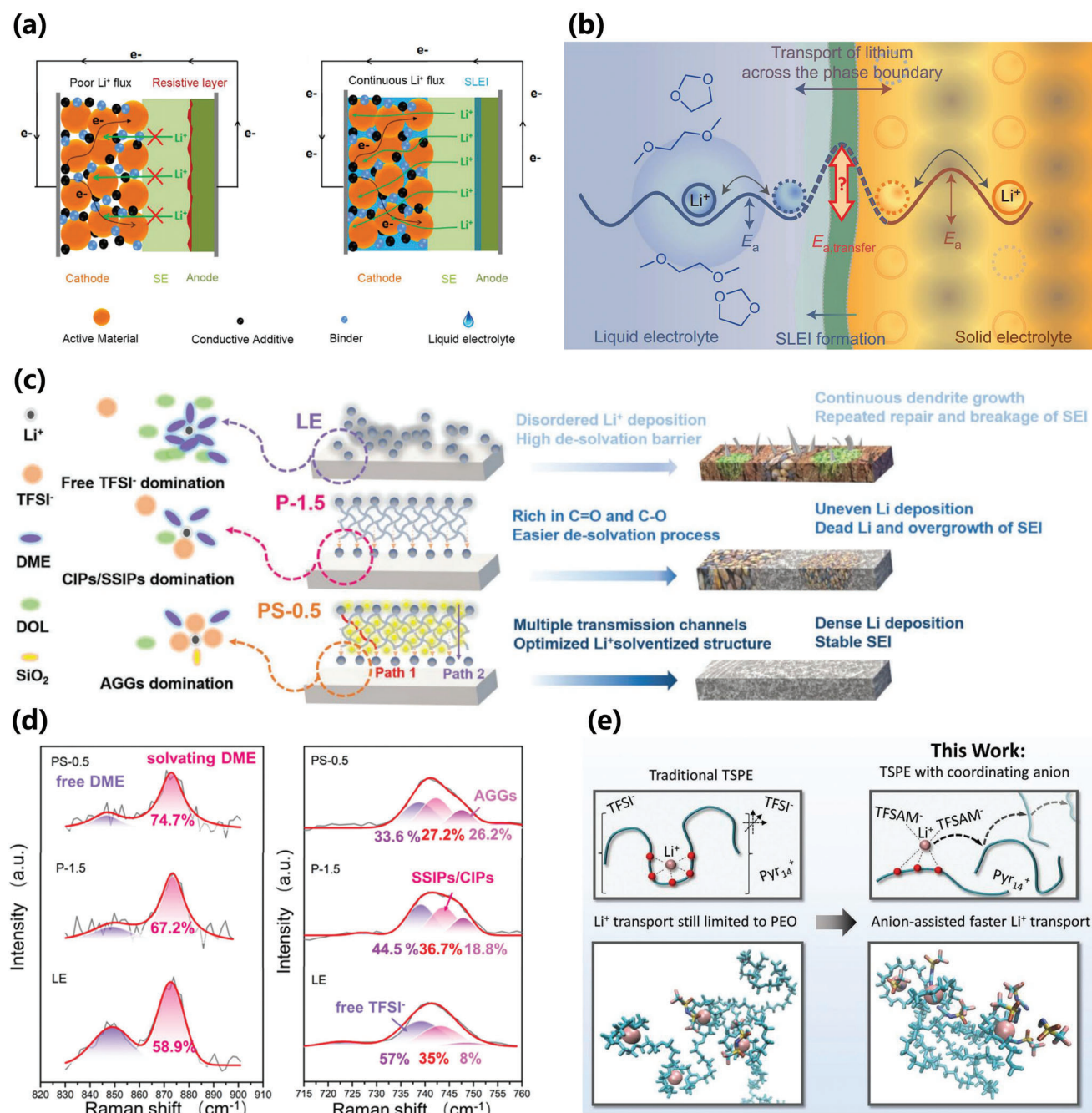


Figure 4. a) Li^+ conduction in the SSLIBs using CLEs. Reproduced with permission.^[49] Copyright 2018, Elsevier. b) The determining step of Li^+ conduction in CLEs. Reproduced with permission.^[50] Copyright 2016, Springer Nature. c) The solvation sheath structures in LE, GPE (P-1.5), and CGPE (PS-0.5) and their influences on Li^+ conduction as well as the SEI formation. Reproduced with permission.^[16] Copyright 2023, John Wiley and Sons. d) The ratio of LE, GPE, and CGPE components. Reproduced with permission.^[16] Copyright 2023, John Wiley and Sons. e) Anion-assisted Li^+ conduction mechanism in GPE with coordinating anion-rich TFSA-based IL. Reproduced with permission.^[59] Copyright 2023, John Wiley and Sons.

2.2. Multi-Component Electrolytes

2.2.1. Ceramic-Liquid Electrolytes

To improve the interfacial contact between the ICEs and electrodes, a certain amount (usually a few μL) of LEs is often added at the interfaces, in which way CLEs can be prepared.^[48] Specif-

ically, a solid-liquid electrolyte interphase (SLEI) will be formed at the interface between the different phases of electrolytes via the electrochemical reactions (Figure 4a). With the assistance of the SLEIs, the degradation of NASICON-type oxide ICEs can be avoided when in contact with the lithium metal anodes.^[49] However, the CLEs can not address the inherent poor flexibility of the electrolyte. Moreover, the addition of the liquid phases may

reduce the overall electrochemical stability, thermal stability, and mechanical properties of the electrolyte.

Inside the CLEs, Li^+ may migrate via complicated pathways: 1) inside the LE component, Li^+ can move with the solvation sheaths; 2) at the SLEIs, Li^+ undergoes de-solvation/re-solvation from the solvation sheath; 3) within the ICEs, Li^+ migrate via the point defects. Due to the minimal amount of liquid phase, ICEs and SLEIs play the dominant role in Li^+ conduction. As shown in Figure 4b, the de-solvation of Li^+ at the SLEIs needs to overcome a significant energy barrier, thus determining the overall Li^+ conduction rate.^[50] As for CLEs with ILs, the energy barrier is greatly influenced by the coordination with the anions of the ILs.^[51]

2.2.2. Gel Polymer Electrolytes

To enhance the ionic conductivity of SPEs, the incorporation of liquid-phase plasticizers is a common approach, and GPEs (also known as quasi-solid electrolytes) can be prepared. In addition to the traditional ion conductive plasticizers like LEs and ILs, other functional solvents such as liquid phase flame retardants, can also serve as potential plasticizers, offering unique non-flammable characteristics to GPEs. For instance, a widely utilized flame-retardant plasticizer, triethyl phosphate (TEP), is frequently employed in the fabrication of non-flammable GPEs.^[52,53] Among the various types of GPEs, those employing LEs as plasticizers have received significant attention due to the combined advantages of LEs and SPEs. The polymer matrices in GPEs enhance their mechanical properties and provide a framework to immobilize the solvent molecules of LEs and therefore preventing undesired side reactions at the electrode/electrolyte interfaces. The addition of liquid phases in GPEs not only improves its ionic conductivity but also facilitates good penetration of the thick electrodes. Nevertheless, it is important to note this enhancement in performance may come at the expense of compromised electrochemical/thermal stability, as well as reduced mechanical strength of the electrolyte.^[54]

In GPEs, the presence of the polymer skeleton provides structural support for the electrolyte, while the plasticizer occupies the interstitial spaces within the skeleton. This leads to the disruption of polymer chain cohesion and the subsequent polymer swelling, thereby increasing the free volume of the polymer. Consequently, the segmental mobility of the polymer is enhanced.^[55,56] Furthermore, the incorporation of plasticizers with high-dielectric constants facilitates the dissociation of the lithium salts.^[57] These synergistic effects of enhanced segment mobility and increased charge carrier concentration further enhance the ion conduction of the polymer matrices. On the other hand, the addition of ion-conductive liquid phase plasticizers can provide additional fast pathways for Li^+ conduction within the GPEs. Therefore, in the LE-added GPEs, Li^+ conduction primarily occurs through two pathways: 1) within the swollen polymer matrices and 2) within the added LEs. The addition of the LEs will generally increase the ionic conductivity of the GPEs, and the conductivity is highly dependent on the amount of LEs. When a small amount of LE is added, the Li^+ conduction in GPEs is mainly associated with the coupling/decoupling within the polymer matrices. However, as the amount of LEs increases, a large fraction of Li^+ will be sol-

vated by the solvent molecules, enabling faster Li^+ conduction within the liquid phase. Notably, in this case, the de-solvation energy barrier of Li^+ at the interfaces remains a critical factor in determining the overall Li^+ conduction rate. In addition, the polymer matrices will positively impact the Li^+ conduction even within the pathway provided by the LEs. For instance, the Raman spectra indicated amount of free solvent molecules was reduced while solvent separated-ion pairs (SSIP)/contact ion pairs (CIP) were increased in the pentaerythritol tetraacrylate (PETEA)/1,2-dimethoxyethane (DME)/1,3-dioxolane (DOL) GPEs compared to the DME/DOL LEs (the cases of P-1.5 and LE in Figure 4c,d).^[16] This reveals the rigid nature of the polymer matrices can immobilize the solvent molecules surrounding the Li^+ , allowing the change to SSIP/CIP-dominated solvation sheaths. Consequently, the de-solvation energy barrier of Li^+ can be reduced, leading to an overall increase in the ionic conductivity of the GPEs.

As for the GPEs with ILs as plasticizers (so-called ionic gels), ILs can also reduce the crystallinity of the polymer matrices and improve the ionic conductivity of the GPEs.^[54] In addition to conducting Li^+ by the ILs themselves, they also affect the conduction of Li^+ in the polymer matrices positively or negatively. For example, the cations of the ILs may be absorbed by the polymer chain and occupy the original Li^+ hopping sites within the polymer.^[58] On the other hand, the addition of some ILs, e.g., trifluoromethanesulfonyl-*N*-cyanoamide (TFSAM), can loosen the coordination between the Li^+ and the polymer chain, thus enhancing the inter-chain conduction and improving the Li^+ transference number (t_{Li^+}) of the GPEs, as illustrated in Figure 4e.^[59]

2.2.3. Composite Polymer Electrolytes

As discussed above, the distinct advantages and disadvantages of SPEs and ICEs can complement each other. Therefore, the ICE fillers can be added to the SPE matrices to form CPEs, demonstrating comprehensive properties.^[11,13,60] Except for ICE fillers (active ceramic fillers), some non-conductive fillers (inert ceramic fillers) can also be introduced to modify the SPEs. The Li^+ conduction mechanism of CPEs varies with the addition of different types of ceramic fillers.^[11,13] Improved ionic conductivity can be achieved in the CPEs due to two main aspects: 1) decreased crystallinity of the polymer matrices and 2) the additional Li^+ conduction pathways provided by the ceramic fillers (only for active fillers) and the polymer-ceramic interfaces (applicable to both types of fillers).

Within the CPEs with inert fillers, Li^+ mainly migrates via 1) intra-polymer and 2) polymer-ceramic interface pathways. As known, the Lewis acidic groups (-OH) on the surface of the acidic ceramic fillers can interact with the anions and polymers via the transient hydrogen bonding, therefore facilitating the dissociation of the lithium salts and reducing the crystallinity of the local polymers. Furthermore, a higher Li^+ concentration can be generated at the polymer-ceramic interface, further improving the ionic conductivity and t_{Li^+} of the CPEs.^[60–62] For the CPEs with basic fillers, the alkaline oxygen on the filler surface can form transient coordination with Li^+ , providing additional sites for Li^+ migration.^[62,63] For the CPEs with neutral fillers, both of the above two mechanisms may exist, but the improvement of

ionic conductivity and t_{Li^+} of the CPEs is limited due to the lower amount of -OH groups and the alkaline oxygens from the fillers. Also, the randomly distributed -OH and alkaline oxygens may also bring the possibility of a stronger bonding between cations and anions, thus reducing the mobility of cations.^[63]

As for the active fillers incorporated CPEs, Li^+ can migrate via three pathways including 1) intra-polymer, 2) intra-ceramic, and 3) polymer-ceramic interfaces. Similar to the inert fillers, the active fillers can also interact with the polymer chains and reduce the crystallinity of the polymer matrices thus enhancing their ion conduction capability. The Lewis acidic groups from the ceramic fillers can facilitate the dissociation of lithium salts and the vacancies on the surface of ceramic fillers enable the hopping of Li^+ , establishing fast Li^+ conduction at the polymer-ceramic interface.^[64]

In the CPEs with active fillers, different pathways play synergistic and competitive roles, and which pathway dominates the Li^+ conduction depends on many factors. In the PEO-lithium bis(trifluoromethanesulfonyl)imide (LiTFSI)/LLZO CPEs, when the filler loading was low (5 wt.%), the Li^+ still preferred to conduct through the intra-polymer pathway. With the filler loading growing, a percolation network would form and lead to a shift of the majority pathway from intra-polymer to intra-ceramic.^[65] The CPE structure might also change from “ceramic-in-polymer” to “polymer-in-ceramic” (Figure 5a).^[66] However, severe agglomeration could occur when higher amounts of ceramic particles are incorporated.^[67] Differently, in another case of PEO- LiClO_4 /LATP CPEs, the maximum ionic conductivity appeared when the loading of LATP was 10 wt.%. The direct contact between LATP particles was yet achieved, and the percolation network across the interface could be formed with lower filler loading. This is because the polymer-ceramic interface pathway dominated the Li^+ conduction with a high-conduction rate.^[68] According to the comprehensive atomic force microscopy analysis on the PEO-sodium bis(trifluoromethylsulfonyl)imide/ $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ (NZSP) CPE, the polymer-ceramic interface was found to offer the fastest pathway for Li^+ conduction.^[69]

The dominant pathway for Li^+ conduction in CPEs varies in different cases, but the polymer-ceramic interface is generally accepted as a fast and important pathway. Thus, the morphology of the fillers can greatly influence the ionic conductivity of CPEs, as it directly determines the area of the polymer-ceramic interface. As illustrated in Figure 5b, with the particle size of garnet-type $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$ (LLZTO) fillers decreased in the PEO-LiTFSI/LLZTO CPEs, the percolation threshold also decreased, while the ionic conductivity showed obvious increment.^[70] This could be attributed to the larger specific surface area of smaller particles. Moreover, the transition from nanoparticles to nanowires/nanofibers or even ceramic frameworks could also enlarge the specific surface area of fillers (Figure 5c), enabling the formation of a continuous polymer-ceramic interface for Li^+ conduction (Figure 5d).^[71–73] The PAN- LiClO_4 /LLTO nanowires CPE demonstrated a significant increase in ionic conductivity, in comparison to the PAN- LiClO_4 /LLTO nanoparticles CPE.^[72] The CPE could even be prepared by injecting PEO-LiTFSI polymer into the pre-formed LLZO framework, enabling a continuous and interconnected ceramic/polymer interface.^[74,75] To further improve the ionic con-

ductivity, aligning the nanowires along the Li^+ conduction direction (Figure 5e) might be an effective strategy.^[73] The PAN- LiClO_4 /aligned LLTO nanowires CPE exhibited an order of magnitude higher ionic conductivity than the one with randomly distributed nanowires.

2.2.4. Composite Gel Polymer Electrolytes

To achieve the SSEs with more attractive properties, CGPEs can be prepared by combining polymer matrices, ceramic fillers, and liquid-phase plasticizers. The well-designed CGPEs demonstrate outstanding ionic conductivity, superior electrode wetting, and satisfied thermal/electrochemical/mechanical stability. However, due to the intricate composition of CGPEs, uncertainty has been introduced into such electrolyte systems. The addition of liquid phase components may introduce inherent issues associated with liquid additives, including undesired side reactions, corrosion, and safety concerns related to flammability and explosibility. Thus, the liquid-phase plasticizers should be carefully selected. For example, the PVDF-HFP-LiTFSI/LLZO-Ga/TEP-fluoroethylene carbonate (FEC) CGPE showed flame-retardant ability with the assistance of the TEP plasticizer.^[76] The addition of FEC was to improve the interfacial stability between the CGPE and the lithium metal anodes.

The Li^+ conduction mechanism of CGPEs can be elucidated based on that of the GPEs and CPEs. Based on the discussion in Section 2.2.2, the incorporation of SiO_2 fillers into the PETEA-LiTFSI/DME/DOL GPE could assist the polymer matrix to immobilize more solvent molecules and favor the interaction between Li^+ and TFSI⁻, thus facilitating the formation of anionic aggregates (AGGs)-dominated solvation sheaths. (the case of PS-0.5 in Figure 4c,d), in which way the de-solvation energy barrier of Li^+ was further reduced.^[16] Density functional theory (DFT) was utilized to study the interaction between components of PEO-LiTFSI/LLZTO/tetrabutylphosphonium 2-hydroxypyridine (TBPHP) CGPE.^[77] The negative value of the binding energy (E_b) represents the chemical bonding stability between the cations and anions, and the calculation results showed strong coordination between the Li^+ and HP^- , which assisted the dissociation of LiTFSI and facilitated the active LLZTO fillers to provide more mobile Li^+ for migration. Furthermore, the interfaces among PEO, LLZTO, and TBPHP offered several rapid pathways for Li^+ migration, as shown in Figure 5f.

The mechanism of Li^+ conduction in CGPEs is a complex interplay of multiple components. It encompasses not only the individual Li^+ conduction ability of each component but also the intricate interactions between these components and the potential synergistic effects arising from their combination. While the current studies have provided valuable insights into the Li^+ conduction mechanisms in CGPEs, a comprehensive understanding of these mechanisms has yet to be fully elucidated. Therefore, further investigations are desired to investigate the intricate details of Li^+ conduction in CGPEs.

3. Polymer-Based SSEs

SPEs offer superior electrode/electrolyte interfacial contact due to their flexible and soft nature. The limitations of SPEs, such as

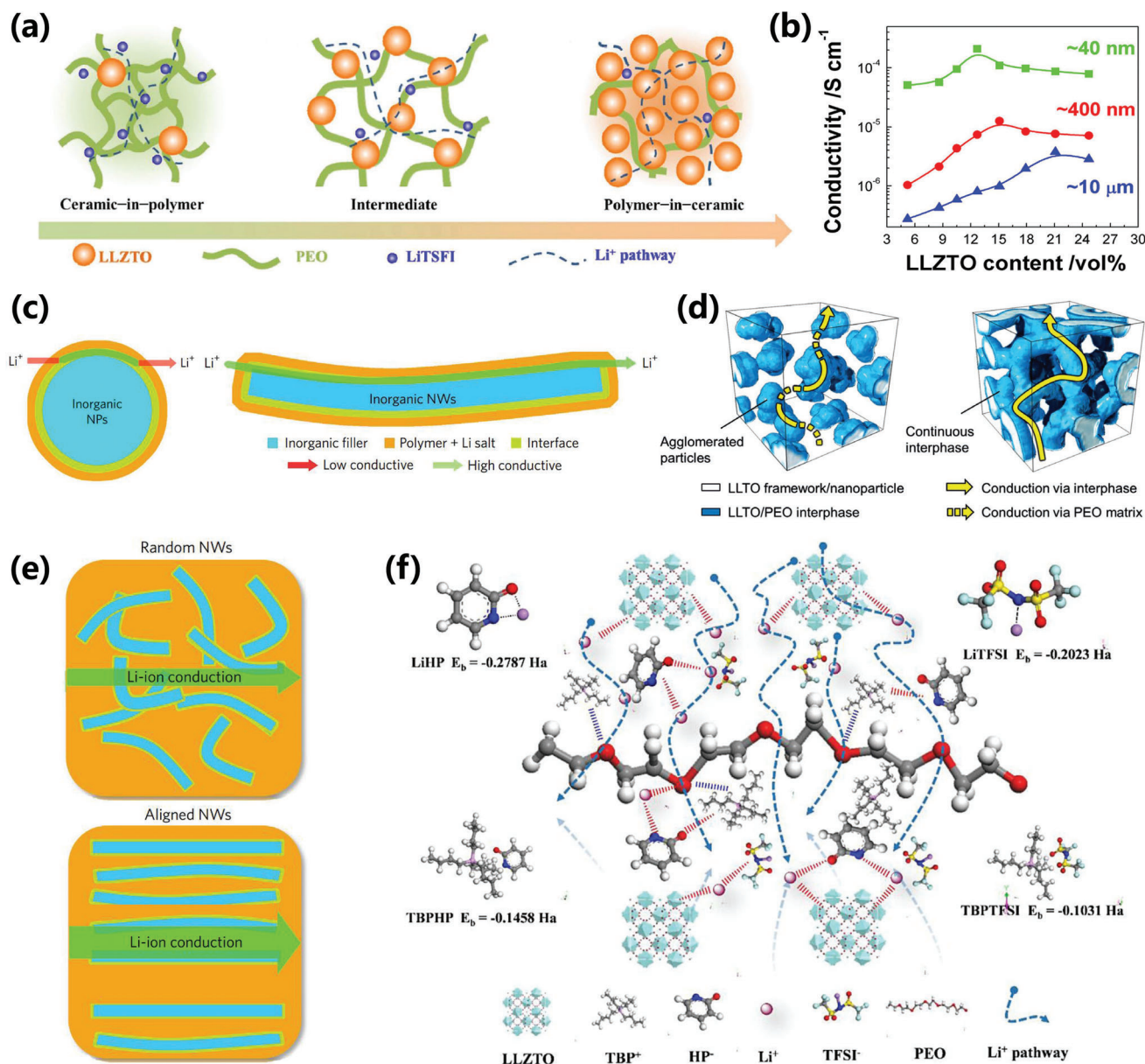


Figure 5. a) Changing from the “ceramic-in-polymer” CPE to “polymer-in-ceramic”. CPE Reproduced with permission.^[66] Copyright 2018, Elsevier. b) Ionic conductivity of the PEO-LiTFSI/LLZTO CPEs with different loadings and various sizes of LLZTO particles. Reproduced with permission.^[70] Copyright 2016, Elsevier. c) The fast Li⁺ migration pathway provided by the polymer-nanowire interface. Reproduced with permission.^[73] Copyright 2017, Springer Nature. d) Continuous polymer-ceramic interface in CPE with nanofiber framework. Reproduced with permission.^[71] Copyright 2018, John Wiley and Sons. e) Aligned nanowires along the Li⁺ conduction direction. Reproduced with permission.^[73] Copyright 2017, Springer Nature. f) Li⁺ conduction pathways in PEO-LiTFSI/LLZTO/THBHP CGPE. Reproduced with permission.^[77] Copyright 2019, Elsevier.

low-room temperature ionic conductivity, narrow electrochemical stability window, and low-thermal stability, have driven the development of GPEs, CPEs, and CGPEs to overcome these shortcomings. Polymer-based SSEs exhibit the most comprehensive properties, making them highly suitable for high-energy-density SSLIBs.

Over several decades of development, PEO, PAN, PVDF, and poly(methyl methacrylate) (PMMA) have emerged as the most widely used polymer matrix materials for polymer-based SSEs. **Figure 6** illustrates their respective properties, which will be fur-

ther discussed in the following sections.

3.1. PEO-Based Polymer Electrolytes

In 1973, PEO was discovered to have the ability to conduct Li⁺.^[78] It should be mentioned that if the molecular weight of the polymer is below 20,000 g mol⁻¹, it is referred to as poly(ethylene glycol) (PEG). PEO exhibits a high-Li⁺ donor number, resulting in appreciable solubility of lithium salts.^[79] Its chain

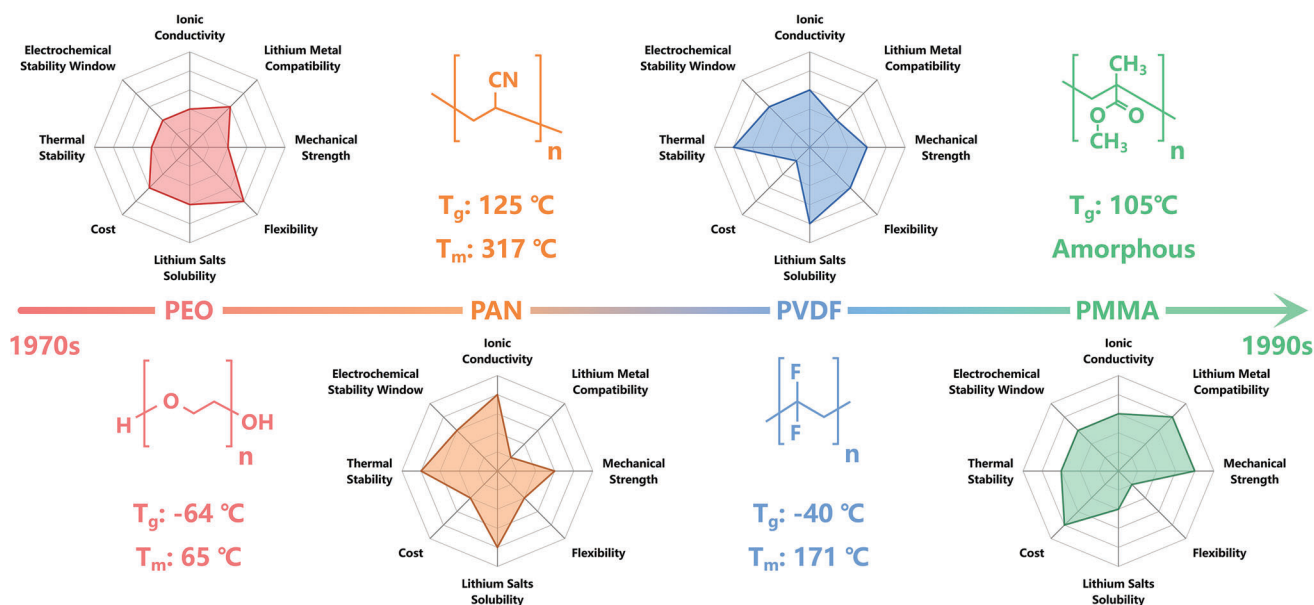


Figure 6. Properties of various polymer matrix materials.

flexibility imparts moderate mechanical properties. However, below its T_m , the crystallinity of PEO is typically 75%–80%, leading to its poor ionic conductivity.^[80] It has been observed that PEO-based electrolytes demonstrate considerably higher ionic conductivity at temperatures above T_m , primarily due to the reduction of their crystallinity. However, at elevated temperatures, its mechanical strength also reduces. Although PEO demonstrates appreciable solubility of lithium salts, its dielectric constant is not very high. Consequently, the low-coordinating anions of lithium salts (e.g., TFSI⁻) remain free in PEO, potentially lowering the t_{Li^+} of the electrolyte. Additionally, the -OH terminal group of PEO chains imposes a relatively narrow electrochemical stability window, which limits its compatibility with high-voltage cathodes.^[27] Also, the gaseous C_2H_4 and H_2 , which have been found to decompose from PEO, will oxidize the lithium metal anodes and cause thick oxide SEI formation.^[81] To address these limitations, several modification strategies have been applied.

Plasticizers with high-dielectric constants can be introduced into the PEO-based electrolytes to enhance ionic conductivity. Das et al. added different plasticizers, such as propylene carbonate (PC), ethylene carbonate (EC), and dimethyl carbonate (DMC) into the PEO-LiClO₄ SPEs to prepare various GPEs.^[82] They found that the dielectric constant of the plasticizer played a crucial role in determining the ionic conductivity of the GPE. The high-dielectric constant plasticizers could facilitate the dissociation of lithium salts or ionic clusters within the PEO matrix, therefore increasing the amount of the mobile Li⁺ and enhancing the ionic conductivity of the resultant GPE. Conversely, the low-dielectric constant plasticizers, like DMC, exhibited the anti-plasticization effect. As discussed in Section 2.2.1, the addition of liquid phases is generally detrimental to the electrochemical stability of the SPEs. Nevertheless, Hassoun et al. reported that the addition of specific plasticizers, such as dimethyl sulfoxide (DMSO) to PEO-based SPE could widen their electrochemical stability window from 3.9 V to 4.1 V (vs Li/Li⁺).^[83]

Adding inert ceramic fillers such as Al₂O₃,^[84,85] SiO₂,^[86,87] TiO₂,^[88] or active ceramic fillers such as LAGP^[89] into the PEO-based SPEs to form CPEs is another way to improve their ionic conductivity and mechanical/electrochemical stability at the same time. As has been discussed in Section 2.2.3, the addition of ceramic fillers can reduce the crystallization of PEO, because the Lewis acidic groups on the surface of those acidic ceramic fillers can interact with the polymer chains via forming the transient hydrogen bond. The reduction of crystallinity of PEO can accelerate its segmental motion, thereby increasing the ionic conductivity of PEO-based electrolytes. The PEO-LiTFSI/LAGP CPE had an ionic conductivity of $6.76 \times 10^{-4} \text{ S cm}^{-1}$, along with an electrochemical stability window of $\approx 5.3 \text{ V}$ (vs Li/Li⁺) at 60 °C. Moreover, the t_{Li^+} was increased from 0.208 to 0.378, and a tensile strength of 0.58 MPa as well as an elongation-at-break of $\approx 80\%$ was achieved. Injecting PEO solvent into the electrospun pre-formed LLZO/PVA nanofiber network could also obtain PEO-based CPE (Figure 7a).^[74] The PEO-LiTFSI/LLZO/PVA nanofiber network CPE exhibited an ionic conductivity of $2.5 \times 10^{-4} \text{ S cm}^{-1}$, with an extremely wide electrochemical stability window of 6.0 V (vs Li/Li⁺) at room temperature. The nanofiber network structure also brought the CPE lower flammability than the one with nanoparticles and contributed to improved lithium dendrite suppression ability.

Other polymers with favorable mechanical properties, such as PAN and PVDF, can be blended with PEO to enhance their mechanical strength.^[90,91] Different polymers can interact with each other, influencing the overall performance of the electrolytes. The specific designs will be discussed in the subsequent sections dedicated to PAN and PVDF-based polymer electrolytes.

It is worth noting that PEO has various derivatives. For instance, the poly(ethylene glycol) dimethyl ether (PEGDME) is one derivative with a wider electrochemical stability window of 4.3 V (vs Li/Li⁺), which replaces the -OH terminal group in PEO with -OCH₃ (Figure 7b).^[27] Poly(ethylene glycol)

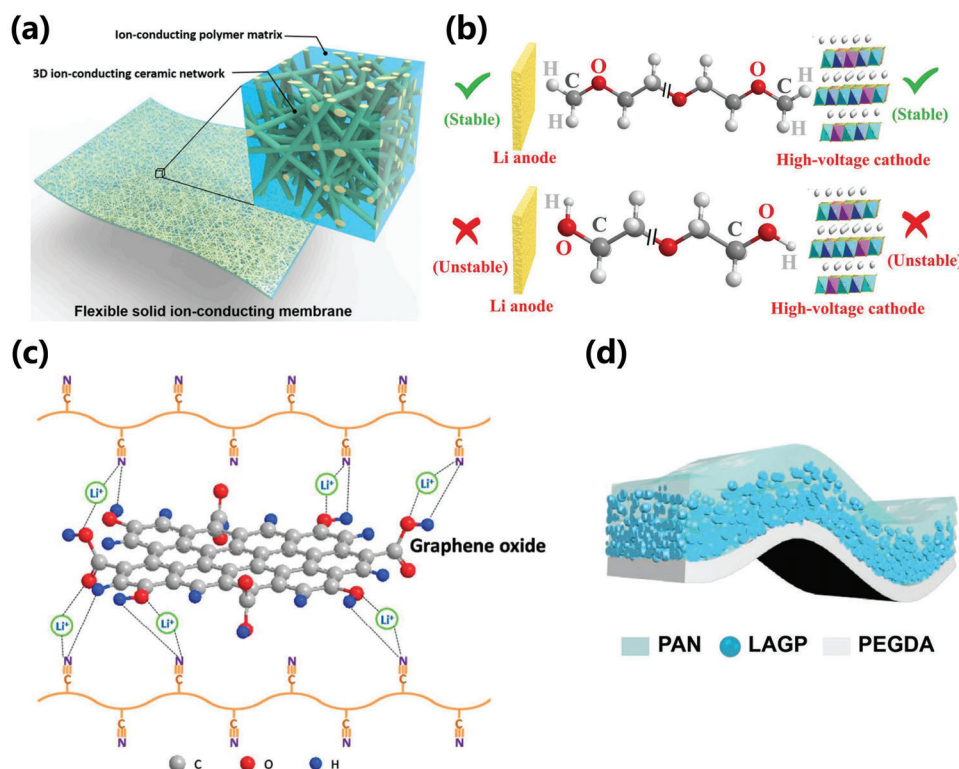


Figure 7. a) Structure of PEO-LiTFSI/LLZO/PVA nanofiber network CPE. Reproduced with permission.^[74] Copyright 2016, the United States National Academy of Sciences. b) The stability between PEO, PEGDME, and the lithium metal anodes/high-voltage cathodes. Reproduced with permission.^[27] Copyright 2020, Royal Society of Chemistry. c) Role of GO fillers in PAN-LiClO₄/GO CPE. Reproduced with permission.^[94] Copyright 2018, Elsevier. d) Asymmetric structure of PAN-LiTFSI/LAGP CPE. Reproduced with permission.^[98] Copyright 2019, John Wiley and Sons.

methacrylate (PEGMA), poly(ethylene glycol) diacrylate (PEGDA) and poly(ethylene glycol) dimethacrylate (PEGDMA) are other typical PEO derivatives that can be in situ polymerized via UV or thermal curing, offering the potential for improving the cathode/electrolyte interfacial contact. The detailed strategies for in situ polymerization will be discussed later.

3.2. PAN-Based Polymer Electrolytes

PAN was initially investigated as the material for polymer electrolytes by Perche et al. in 1975.^[92] It possesses unique polar $\text{—C}\equiv\text{N}$ groups within its polymer chains, which provide pathways for Li⁺ conduction, resulting in higher ionic conductivity at room temperature compared to PEO.^[72,93] The presence of the rigid $\text{—C}\equiv\text{N}$ bonds restricts the rotation of the polymer chains, thus imparting good mechanical strength but brittleness to the PAN-based electrolytes.^[94] In addition, PAN exhibits favorable thermal/electrochemical stability. Consequently, PAN is predominantly employed as the polymer framework in GPEs or blended with other soft polymers to form copolymers, leveraging these desirable properties. However, despite PAN having a relatively wide electrochemical stability window, severe passivation may occur at the PAN/lithium metal anode interface due to their electron receptive properties and strong reducibility of lithium metal. As a result, the $\text{—C}\equiv\text{N}$ groups undergo reduction, leading to the formation of a passivation layer enriched with products derived

from the reduction of $\text{—C}\equiv\text{N}$.^[95] This passivation layer can subsequently increase the interfacial resistance and adversely impact the cycle performance of the battery.

Wang et al. successfully enhanced the performance of PAN-based electrolytes by incorporating graphene oxide (GO) fillers into the polymer matrix.^[94] The functional groups present in GO, such as —OH , —COOH , —C=O , and —COC , effectively mitigated the polarity of the $\text{—C}\equiv\text{N}$ group in PAN chains, thereby enhancing the flexibility of the PAN matrix and improving its compatibility with lithium metal anodes. As depicted in Figure 7c, the additional oxygen atoms provided by GO could coordinate with the Li⁺, facilitating the dissociation of LiClO₄. The resulting PAN-LiClO₄/GO CPE exhibited an improved ionic conductivity of $4.0 \times 10^{-4} \text{ S cm}^{-1}$ at 30 °C, with an electrochemical stability window of 4.3 V (vs Li/Li⁺). Cui et al. reported that the ionic conductivity of PAN-LiClO₄/LLTO CPE with the LLTO nanowires aligned toward the Li⁺ conduction direction at 30 °C was $6.05 \times 10^{-5} \text{ S cm}^{-1}$, and the nanowires brought the CPE long-term structural stability. Ahn et al. fabricated the PAN-based CGPE by immersing the EC/DMC-LiPF₆ LE into the electrospinning PAN/BaTiO₃ (BTO)/SiO₂/Al₂O₃ porous membrane.^[96] Compared to the PAN/EC/DMC-LiPF₆ GPE, the PAN/Al₂O₃/EC/DMC-LiPF₆ CGPE exhibited superior performance, including higher room temperature ionic conductivity ($3.46 \times 10^{-3} \text{ S cm}^{-1}$ vs $1.21 \times 10^{-3} \text{ S cm}^{-1}$), wider electrochemical stability window (5.7 V vs 5.2 V) and lower electrode/electrolyte interfacial resistance (180 Ω vs 624 Ω). Hou et al. proposed

a PAN/PEO cross-linked network with improved mechanical strength compared to PEO.^[90] Due to the strong interaction between the $\text{—C}\equiv\text{N}$ group in PAN and the EC/DMC/diethyl carbonate (DEC)-LiPF₆ LE plasticizer, the resulting GPE exhibited a wide electrochemical stability window of 5.5 V (vs Li/Li⁺). Furthermore, the incorporation of PAN lowered the crystallinity of PEO. Coupled with the substantial LE uptake ability of the copolymer, the GPE exhibited an excellent ionic conductivity of $3.0 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature. Pan et al. developed a GPE with biodegradable PAN/polylactic acid-block-ethylene glycol (PALE) polymer skeleton and DOL/DME-LiTFSI LE.^[97] The functional groups ($\text{—C}\equiv\text{N}$, —C=O , and —COC) on the PALE chains aided in the dissociation of LiTFSI, leading to an ionic conductivity of $8.4 \times 10^{-4} \text{ S cm}^{-1}$ and an electrochemical stability window of 5.11 V (vs Li/Li⁺) at room temperature. Moreover, the highly cross-linked polymer skeleton exhibited flame-retardant properties during combustion. To prevent direct contact between PAN and lithium metal anodes, special structural designs of the electrolytes was employed. Guo et al. designed a Janus structure PAN-LiTFSI/LAGP CPE (Figure 7d), with PAN contacting the cathode side to achieve good interfacial contact and LAGP contacting the anode side to suppress the lithium dendrites and the potential side reactions.^[98] By introducing an in situ polymerized PEGDA thin layer between the lithium metal anode and LAGP-side electrolyte, the electrode wetting could be further enhanced. Other approaches involving lithium-stable polymers will be introduced below.

3.3. PVDF-Based Polymer Electrolytes

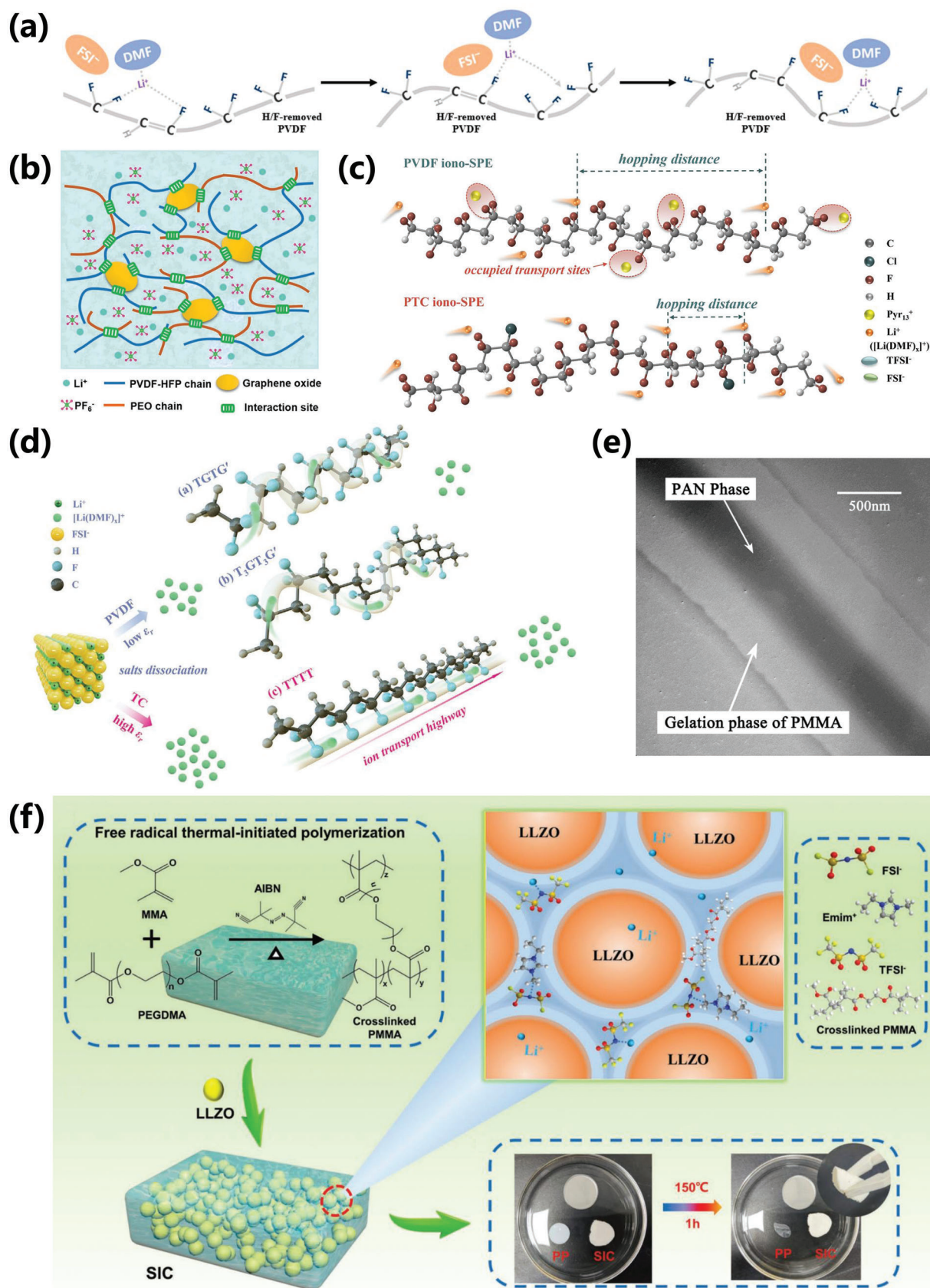
PVDF was studied in 1975 together with PAN as the material for polymer electrolytes.^[92] Like PAN, PVDF exhibits excellent thermal/electrochemical stability and high-mechanical strength. The presence of the —CF functional group in PVDF grants it a high-dielectric constant and strong electron absorption properties, facilitating the efficient dissociation of lithium salts and enabling a high-charge carrier concentration. Similar to PEO, PVDF demonstrates a relatively high-crystallinity of $\approx 65\%$ – 78% at room temperature.^[99] Even though the fluorinated polymers are unstable to lithium metal anodes, a desirable LiF-rich SEI can be formed at the interface between the PVDF-based electrolytes and the lithium metal anode, facilitating the interfacial ion conduction and improving the interfacial mechanical strength as well.

N,N-dimethylformamide (DMF) is commonly employed as a solvent for preparing PVDF-based electrolytes and can also serve as a plasticizer for PVDF-based GPEs. As depicted in Figure 8a, residual DMF solvents remaining after the preparation of PVDF-based electrolytes can solvate with Li⁺, forming $[\text{Li}(\text{DMF})_x]^+$ complexes and conducting Li⁺ by coordinating/uncoordinating with fluorine atoms on the PVDF chains.^[100,101] Yang et al. utilized DMF as both the solvent and plasticizer to fabricate PVDF-based GPE, achieving an ionic conductivity of $1.2 \times 10^{-4} \text{ S cm}^{-1}$ at room temperature.^[102] However, the addition of excessive liquid compromised the mechanical stabilities of PVDF and increased the safety risk. To address this issue, palygorskite $[(\text{Mg},\text{Al})_2\text{Si}_4\text{O}_{10}(\text{OH})]$ nanowires were incorporated into the aforementioned GPE to form a CGPE. The interconnected

network of the nanowires improved the mechanical strength as well as the ionic conductivity ($1.4 \times 10^{-4} \text{ S cm}^{-1}$ at room temperature) of the CGPE.

PVDF is often combined with other polymers to simultaneously leverage the favorable mechanical properties of PVDF and also improve its ionic conductivity. Yang et al. Prepared the electrospun PVDF/PEO nanofibrous membrane with enhanced mechanical properties due to the support of the PVDF framework.^[91] Within the membrane, the PVDF and PEO chains were tangled and their functional groups could interact with each other, thereby destroying the orderly arrangement of polymer chains and decreasing the crystallinity of both polymers. Moreover, the microporous structure also facilitated the efficient capture of LE, due to the similar polarity and chemical structure between PEO and the absorbed LE. Thus, the PVDF/PEO/EC/PC/DMC-LiPF₆ GPE exhibited an ionic conductivity of $4.8 \times 10^{-3} \text{ S cm}^{-1}$ and an electrochemical stability window up to 4.8 V (vs Li/Li⁺) at room temperature.

One widely used PVDF-based copolymer is poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), which comprises symmetric VDF monomers and asymmetric HFP monomers. PVDF is recognized as a crystal polymer, while HFP is amorphous. The bulky —CF_3 groups of HFP greatly reduce the overall crystallinity and increase the flexibility of PVDF-HFP compared to pure PVDF. In addition, the increase in the fluorine content enhances its hydrophobicity.^[103] The coexistence of these two polymers in PVDF-HFP enables both the desirable ionic conductivity and mechanical strength. Saika and Kuma compared the ionic conductivity of PVDF and PVDF-HFP-based GPEs using the same plasticizer (PC/DEC).^[104] The results showed that the PVDF-HFP-based GPE exhibited superior ionic conductivity ($7.5 \times 10^{-3} \text{ S cm}^{-1}$) compared to the PVDF-based GPE ($1.3 \times 10^{-3} \text{ S cm}^{-1}$). This improvement could be attributed to the presence of randomly mixed VDF and HFP units, which provided more amorphous regions for Li⁺ conduction. Tang et al. fabricated PVDF-HFP/PEO/GO/EMC-LiPF₆ CGPE through weak hydrogen bonding interactions.^[105] The presence of hydrogen bonds among the PVDF-HFP, PEO, and GO disrupted the ordering of the polymer chains, favoring the formation of a 3D porous network (Figure 8b). Thus, the ionic conductivity of the resulting CGPE reached $2.1 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature. Wang et al. incorporated hydroxyapatite (HAP) fillers and *N*-methyl pyrrolidone (NMP) plasticizer into the PVDF-HFP matrix to prepare a CGPE.^[106] The addition of HAP fillers and NMP plasticizer facilitated the dissociation of LiTFSI and promoted the conduction of Li⁺, leading to an ionic conductivity of $7.4 \times 10^{-4} \text{ S cm}^{-1}$ at room temperature. Hydrogen bonds were formed between the —OH groups of HAP and the fluorine atoms of the PVDF-HFP matrix, thereby enhancing the mechanical strength of the CGPE, whose tensile strength was 3.89 MPa and elongation-at-break was 403.3%. Additionally, the CGPE also showed self-extinguish properties, which could avoid serious fire risks and keep its original shape. Kim et al. electrospun the PVDF-HFP/Al₂O₃ composite polymer matrix and further prepared the PVDF-HFP/Al₂O₃/EC/DMC-LiPF₆ CGPE.^[107] The CGPE demonstrated improved ionic conductivity and electrochemical stability of $4.1 \times 10^{-3} \text{ S cm}^{-1}$ and 5.0 V (vs Li/Li⁺), superior over the PVDF-HFP/EC/DMC-LiPF₆ GPE ($3.0 \times 10^{-3} \text{ S cm}^{-1}$ and 4.6 V vs Li/Li⁺). This was because the high-dielectric



constant of Al_2O_3 particles could couple with the absorbed LE (e.g., $\text{Al}_2\text{O}_3\text{-PF}_6^-$ and $\text{Al}_2\text{O}_3\text{-EC}$), and the coordination between Al_2O_3 and the Li^+ could further promote the Li^+ migration.

Poly(vinylidene fluoride-co-trifluoroethylene) (PVDF-TrFE) is another important copolymer based on PVDF, which can be synthesized through the polymerization of VDF and TrFE monomers. The incorporation of TrFE enhances the crystallinity of the polymer. Unlike pure PVDF, PVDF-TrFE can undergo a solid-state ferroelectric to paraelectric phase transition.^[103] Another new terpolymer, poly(vinylidene fluoride-co-trifluoroethylene-co-chlorotrifluoroethylene) [P(VDF-TrFE-CTFE)], has recently been synthesized by incorporating PVDF-TrFE with CTFE. P(VDF-TrFE-CTFE) demonstrates relaxor ferroelectric behavior, along with enhanced dielectric constant and desirable polarity characteristics. Huang et al. reported a P(VDF-TrFE-CTFE)-based GPE with more balanced Li^+ conduction in different phases.^[58] *N*-Propyl-*N*-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide ($\text{Pyr}_{13}\text{TFSI}$) IL were added to the P(VDF-TrFE-CTFE) framework, and the moderate polarity of P(VDF-TrFE-CTFE) reduced the occupation of the cations of the IL at the Li^+ hopping site of the polymer chain, as shown in Figure 8c. Combined with the improved dissociation of Li^+ -anions clusters due to the high-dielectric constant of P(VDF-TrFE-CTFE), the overall ionic conductivity of the GPE reached $5.75 \times 10^{-4} \text{ S cm}^{-1}$ at 25°C . He et al. added the PVDF-TrFE copolymer into the P(VDF-TrFE-CTFE) terpolymers to further disturb the crystallization of the polymers. In this way, the conformation of fluorine atoms changed from the TGTG'-T₃GT₃G' mixed to the TTTT conformation, as illustrated in Figure 8d.^[108] Since the conformation of fluorine atoms strongly influence the ionic conductivity of PVDF-based electrolytes, the TTTT conformation provided a highway for $[\text{Li}(\text{DMF})_x]^+$ conduction. Consequently, an ionic conductivity of $2.37 \times 10^{-4} \text{ S cm}^{-1}$ and a t_{Li^+} of 0.61 were brought to the GPEs at room temperature.

3.4. PMMA-Based Polymer Electrolytes

PMMA was first studied as a polymer matrix of GPE in 1985.^[109] PMMA is a transparent amorphous polymer and possesses a higher Li^+ conduction ability than PEO at room temperature. The most important feature of PMMA is its good interfacial stability with lithium metal anodes.^[55] PMMA-based electrolytes also have the capacity to contain a large amount of LE plasticizers thanks to their ester groups.^[110] Additionally, PMMA offers the advantage of lower manufacturing costs compared to other polymers, thereby enabling the potential for large-scale production. However, the amorphous structure of PMMA often limits its wide application. Consequently, the incorporation of plasticizers is required to enhance the flexibility of the PMMA-based SPEs or CPEs. On the other hand, the PMMA-based electrolytes can

have close blending with other high-strength polymer matrices and good electrode/electrolyte interfacial contact, as it can be prepared by in situ polymerization of MMA.

Ma et al. synthesized PMMA/polystyrene copolymers with different molar ratios and introduced PC- LiClO_4 as the plasticizer to prepare GPE.^[111] The introduction of polystyrene block improved the mechanical properties, thermal stability, and liquid preservation ability of the GPE. Utilizing the lithium-stable property of PMMA, Sui et al. synthesized a unique PAN (core)-PMMA (shell) nanofiber structure by coaxial electrospinning, effectively avoiding the direct PAN/lithium metal anode contact.^[112] PMMA had a good affinity to LEs, and the scanning electron microscopy image of PAN-swollen PMMA structure after gelation is shown in Figure 8e. Zhang et al. developed an elastomeric electrolyte utilizing PMMA-polyisoprene copolymer as the elastomer matrix and tetrahydrofuran (THF)- LiTFSI for electrode/electrolyte wetting.^[113] The elastomeric electrolyte exhibited a noteworthy ionic conductivity of $4.6 \times 10^{-4} \text{ S cm}^{-1}$ and a wide electrochemical stability window of 5.08 V (vs Li/Li^+). Additionally, the elastomeric electrolyte displayed a high extensibility of $\approx 560\%$, significantly improving the poor flexibility of PMMA. Li et al. prepared a GPE based on an electrospun PAN/PMMA framework and *N*-methyl-*N*-butylpyrrolidinium bis(trifluoromethanesulfonyl)imide ($\text{Pyr}_{14}\text{TFSI}$) IL.^[114] Compared to the PAN framework, the PAN/PMMA framework had a reduced fiber size from 600 to 450 nm and a higher porosity of 86% than 78%. Together with the excellent compatibility with ILs of PMMA, the IL uptake of the PAN/PMMA framework reached 480%. This brought the GPE an ionic conductivity of $3.6 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature. Thanks to the high-electrochemical stability of $\text{Pyr}_{14}\text{TFSI}$, the PAN/PMMA- $\text{Pyr}_{14}\text{TFSI}$ GPE also exhibited electrochemical stability $> 5 \text{ V}$ (vs Li/Li^+). Ma et al. proposed a self-healing GPE design by grafting Emim⁺ IL chain units into the PMMA backbone.^[115] PMMA demonstrated good chemical stability toward lithium metal anodes, while Emim⁺ facilitated the formation of hydrogen bonds with oxygen-containing groups on PMMA chains, enabling a self-sealing property at the anode/electrolyte interface. As a result, the electrolyte could effectively heal the defects and cracks caused by lithium dendrites, bringing excellent cycle performance to the cell.

Sharma et al. prepared a CGPE by adding the PC/DEC plasticizer and SiO_2 nanofibers into the PMMA- LiClO_4 SPE.^[116] The CGPE exhibited an impressive ionic conductivity of $2.56 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature. Furthermore, the addition of nanofibers significantly improved the mechanical strength of the CGPE, resulting in a yield strength of 4.38 MPa and an elongation-at-break of 28%. Song et al. developed a super-concentrated ionogel-in-ceramic CGPE by utilizing cross-linked PMMA as the polymer matrix.^[117] The fabrication process involved thermal curing with MMA monomer and PEGDMA

Figure 8. a) The unique $[\text{Li}(\text{DMF})_x]^+$ conduction mechanism in PVDF with residual DMF solvents. Reproduced with permission.^[101] Copyright 2022, Elsevier. b) Interaction among PVDF-HFP, PEO, and GO fillers in CPE. Reproduced with permission.^[105] Copyright 2018, John Wiley and Sons. c) Shortened Li^+ hopping distance enabled by avoiding the Pyr_{13}^+ occupation at the Li^+ hopping sites in P(VDF-TrFE-CTFE) polymer matrix. Reproduced with permission.^[58] Copyright 2023, John Wiley and Sons. d) Effects of dielectric constants on lithium salt dissociation and fluorine atoms conformations on Li^+ conduction in different polymer matrices. Reproduced with permission.^[108] Copyright 2023, John Wiley and Sons. e) SEM image of PAN (core)-PMMA (shell) nanofiber structure after gelation. Reproduced with permission.^[112] Copyright 2014, Elsevier. f) The synthesis process of the CGPE and their elevated thermal stability. Reproduced with permission.^[117] Copyright 2022, John Wiley and Sons.

crosslinker, as depicted in Figure 8f. The LLZO particles were added as ceramic fillers, and 1-ethyl-3-methylimidazolium bis(fluorosulfonyl)imide (EmimFSI)-LiTFSI IL was immobilized within the polymer matrix. The resulting CGPE exhibited a notable ionic conductivity of $1.33 \times 10^{-3} \text{ S cm}^{-1}$, a high- t_{Li^+} of 0.89 and a wide electrochemical stability window of 5.5 V (vs Li/Li⁺) at 25 °C. Additionally, compared to the commercially used polypropylene separator, the CGPE displayed enhanced thermal stability.

4. Design of Electrode/Polymer-Based SSE Interfaces for High-Energy-Density SSLIBs

Unlike the LEs which can penetrate and wet the thick electrodes with high-mass-loading, even though the flexible polymer-based SSEs can provide good contact with the cathodes, the Li⁺ conduction within the cathodes and at the cathode/electrolyte interfaces in the SSLIBs is usually poor. What was worse, the interfacial space-charge layers further hinder the Li⁺ conduction at the electrode/electrolyte interfaces. Additionally, the electrochemical stability at the cathode/electrolyte interfaces is another significant issue that limits the wide application of the polymer-based SSEs, since the narrow electrochemical stability window of the electrolytes will potentially restrict the utilization of high-voltage cathodes, thereby directly impacting the energy-density of the resultant SSLIBs. Differently, the lithium metal anode/electrolyte interfaces may face the problem of chemical instability. The growth of lithium dendrites also significantly impacts battery safety and cycle performance. Thus, innovative designs targeting the electrode/electrolyte interfaces have been proposed to overcome these challenges.

4.1. Cathode/Polymer-Based SSE Interfaces

4.1.1. Interfacial Contact and Wetting

In comparison to the ICEs, soft and flexible polymer-based SSEs demonstrate superior contact with electrodes. However, when matched with high-mass-loading cathodes, the polymer-based SSEs still encounter challenges in effectively penetrating and wetting the cathodes, resulting in suboptimal Li⁺ conduction within the cathodes.^[118] To address this issue, two possible solutions have been proposed: in situ polymerization and integrated cathode-electrolyte strategy.

In Situ Polymerization: In situ polymerization provides a viable solution to address the electrode contact and wetting problem and therefore enables the establishment of ion conduction pathways within cathodes. In this process, the liquid monomer precursors are usually directly cast onto the cathode or injected into the pre-assembled cells, thus the liquid state precursors can effectively permeate the thick cathodes. From the mechanisms' point of view, the polymerization of monomers can be classified into chain polymerization and stepwise polymerization. The in situ polymerization for polymer-based SSEs is mostly initiator-involved chain polymerization, which can be further divided into free-radical polymerization, cation polymerization, and anion polymerization depending on different active centers.

In situ polymerization of PEO derivatives-based electrolytes is mostly free-radical polymerization, and they can be initiated under different external conditions with specific initiators. For instance, UV curing is a convenient method that only involves UV lights and photoinitiators [e.g., 2-hydroxy-2-methylpropiophenone (HMPP)]. Deng et al. employed this method to polymerize the poly(ethylene glycol) methyl ether methylacrylate (PEGMEMA)/PEGDA-LiTFSI precursor, which was directly cast on the LiMn_{0.8}Fe_{0.2}PO₄ (LMFP) cathode.^[119] The precursor incorporated the HMPP initiator and was cured under UV irradiation. They found that the LMFP (mass-loading: 2.0 mg cm⁻²) || PEGMEMA/PEGDA-LiTFSI SPE (thickness: ≈180 μm) || Li cell showed an initial discharge capacity of 158.8 mAh g⁻¹ and a capacity retention of 84.6% after 240 cycles under 0.2 C at 60 °C. Guo et al. used UV curing to in situ polymerize a GPE on the LFP cathode.^[120] They first prepared a PVDF-HFP/PVDF/Poly(vinylpyrrolidone) porous membrane and used it as a supportive framework. Then they filled the framework with the PEGDA/pentaerythritol tetra-(mercaptopropionate) (PETMP)/2,2,2-trifluoroethyl methacrylate (TFEMA) curable monomers with succinonitrile as plasticizer and LiTFSI as lithium salt, finally in situ polymerized them onto the cathode by UV curing, as depicted in Figure 9a. The obtained LFP (mass-loading: 0.6–0.7 mg cm⁻²) || UV-cured GPE (thickness: ≈24 μm) || Li cell demonstrated attractive performances with an initial discharge capacity of 153 mAh g⁻¹ and a capacity retention of 91% after 150 cycles under 0.1 C at 25 °C. Notably, the UV curing however poses challenges when applied to the pre-assembled cells with anodes, due to the limited irradiated area. Consequently, UV curing is only feasible for polymer precursors directly cast on the cathode, followed by the subsequent assembly of the anode. This limitation restricts the contact between the anode and the electrolyte and may cause cycle instability for the cell. Thermal curing is an alternative method for in situ free-radical polymerization of PEO derivatives-based SSEs in pre-assembled cells, which commonly involves the 2,2-azobisisobutyronitrile (AIBN) initiator. Zhao et al. added 2-cyanoethyl acrylate (CA)/BN with PEGDA cross-linker and AIBN initiator into EC/DEC-LiPF₆ LE to prepare CGPE precursor, then injected the precursor into the pre-assembled cell and cured under 70 °C (Figure 9b).^[121] According to the SEM images, the in situ polymerization brought uniform CGPE wrapping and full penetration of the porous cathode. The LiCoO₂ (LCO, mass-loading: 2.1 mg cm⁻²) || CA/PEGDA/BN/EC/DEC-LiPF₆ CGPE || Li cell demonstrated a discharge capacity of 113.2 mAh g⁻¹ after 200 cycles under 0.1 C at room temperature, with a capacity retention of 77%. Cui et al. in situ polymerized the PEG-LiTFSI/CA/LLZTO CPE with the aid of AIBN under 80 °C.^[122] The interaction between the —C≡N group of CA and LCO cathode contributed to the formation of a —C≡N and LiF-rich CEI, and thus the LCO (mass-loading: 6 mg cm⁻²) || PEG-LiTFSI/CA/LLZTO CPE || Li cell delivered an excellent initial discharge capacity of 156.9 mAh g⁻¹ and capacity retention of 80.7% after 500 cycles at 60 °C and 0.5 C.

The in situ ring-opening polymerization of ring monomers (e.g., DOL) is mostly cation polymerization, in which Lewis acids can be used as initiators. Notably, some complex anions of lithium salts are Lewis acid-stabilized simple anions.^[123] Therefore, in some cases, lithium salts such as Lithium

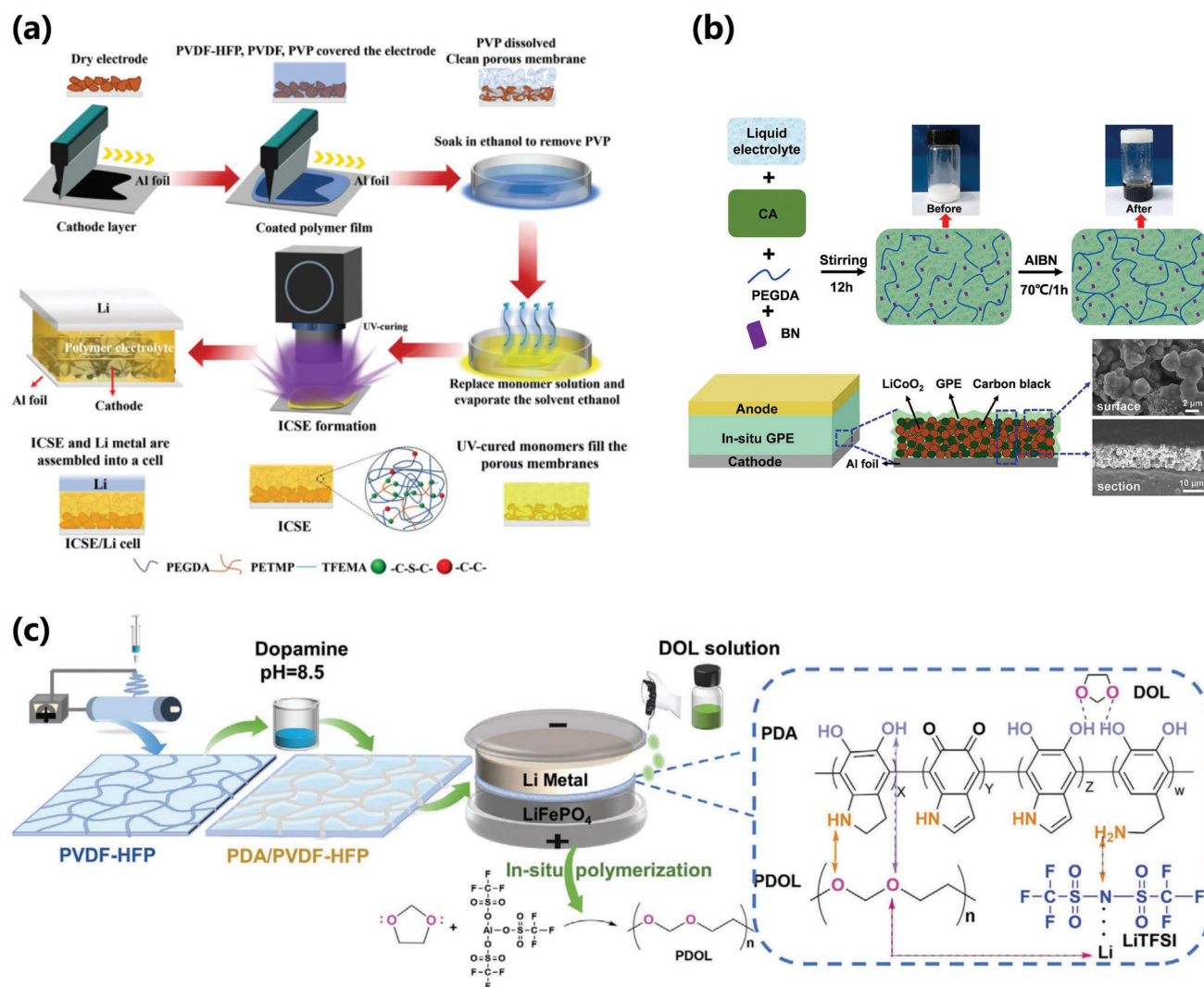


Figure 9. Different in situ polymerization strategies: a) in situ polymerization via UV curing. Reproduced with permission.^[120] Copyright 2021, Royal Society of Chemistry. b) In situ polymerization via thermal curing. Reproduced with permission.^[121] Copyright 2021, Elsevier. c) In situ ring-opening polymerization. Reproduced with permission.^[125] Copyright 2022, John Wiley and Sons.

difluoro(oxalato)borate (LiDFOB) can decompose trace Lewis acid for initiating cation polymerization without extra initiators. Sui et al. developed an ultrathin and flexible CGPE by in situ polymerizing DOL within the inorganic soft SiO₂ nanofibrous (ISN) skeleton.^[124] LiDFOB was incorporated and acted as both lithium salt and initiator source in the electrolyte. The interface charge-transfer resistance of the LFP (mass-loading: 2.4 mg cm⁻²) || PDOL-LiDFOB/ISN CGPE (thickness: 32–34 μm) || Li cell was only 30.8 Ω due to the intimate electrode/electrolyte interfaces carried out by in situ polymerization. After 300 cycles under 0.2 C at room temperature, the cell delivered a high-discharge capacity of 150.9 mAh g⁻¹ with a capacity retention of 97.1%. Even with a higher cathode mass-loading of 11.7 mg cm⁻², the cell still cycled stably under 1 C at 50 °C for over 60 cycles. The same group also successfully fabricated self-enhanced GPE by in situ polymerizing DOL within a polydopamine (PDA)-modified PVDF-HFP nanofibrous backbone using Al(OTf)₃ as initiator, as presented in Figure 9d.^[125] The excellent affinity between the nanofi-

brous membrane and PDOL provided continuous pathways for Li⁺ conduction. Additionally, hydrogen bonds could be formed between PDA and PDOL, thereby enhancing the mechanical strength of the electrolyte. The LFP (mass-loading: 2.2 mg cm⁻²) || PDOL/PDA/PVDF-HFP-LiTFSI GPE (thickness: ≈90 μm) || Li cell remained at a discharge capacity of 144.6 mAh g⁻¹ after 200 cycles under 0.2 C at 25 °C, with extremely high-capacity retention of 96.03%. Remarkably, even when the current density was increased to 2 C, the cell could still stably operate for over 800 cycles.

Integrated Cathode-Electrolyte: The poor Li⁺ conduction within the cathodes presents a challenge in achieving high-energy-density SSLIBs using high-mass-loading cathodes. To address this issue, one approach involves incorporating certain polymer electrolytes into the cathode as both the ion conductors and binders, then casting the polymer-based SSE solvents on the cathodes to fill in the pores. As a result, the same components of the cathodes and the electrolytes can be bonded together to

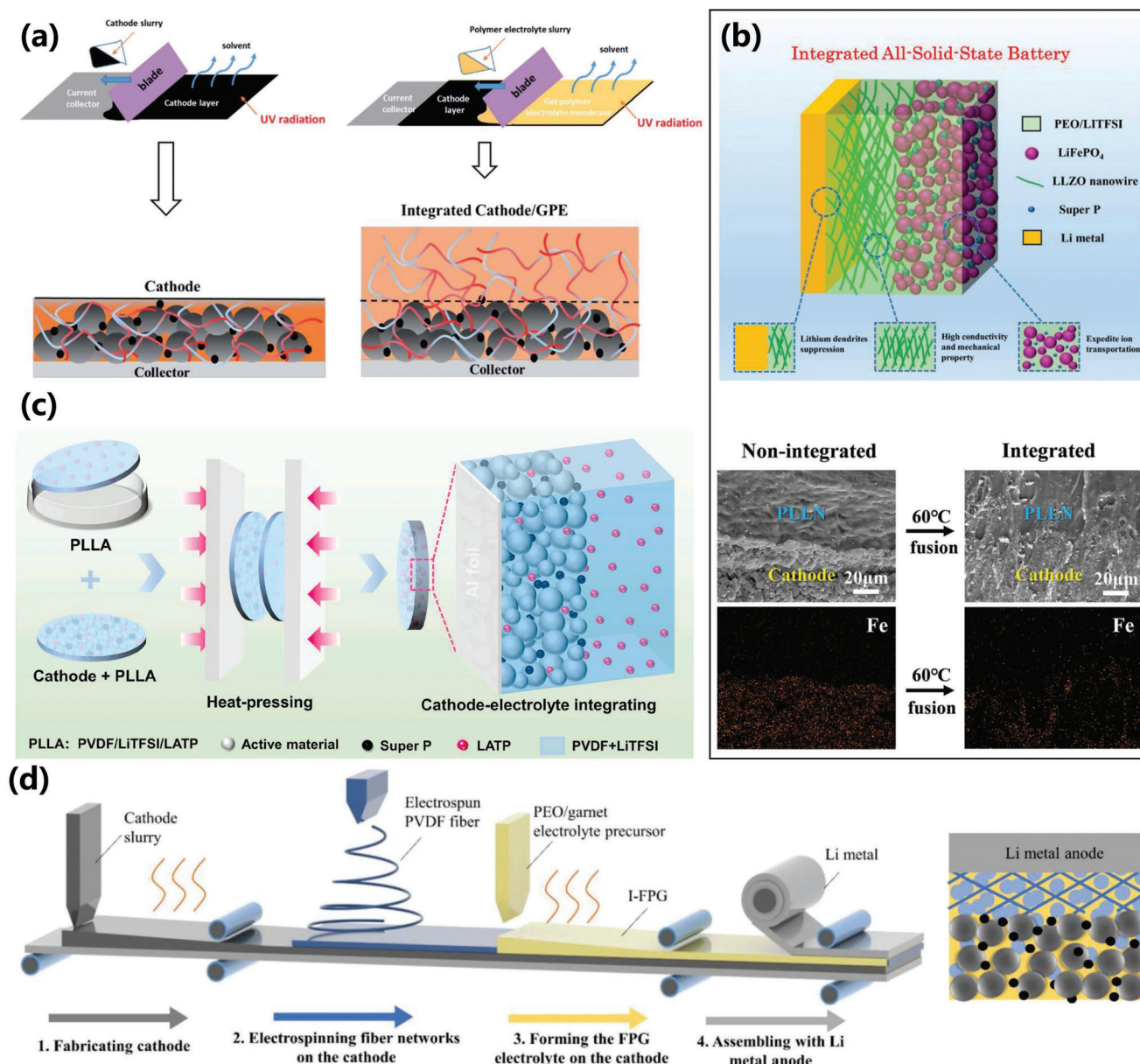


Figure 10. Various methods to prepare the integrated cathode/electrolyte structures: a) Integrated LFP cathode-GelMA/PEGDA-LiTFSI GPE structure by UV curing. Reproduced with permission.^[127] Copyright 2021, Royal Society of Chemistry. b) Integrated LFP cathode-PLLN CPE structure by fusing. Reproduced with permission.^[128] Copyright 2019, John Wiley and Sons. c) Integrated LFP cathode-PLLA CPE structure by hot pressing. Reproduced with permission.^[129] Copyright 2019, Elsevier. d) Integrated LFP cathode-PVDF/FPG CPE fabrication process with electrospinning. Reproduced with permission.^[130] Copyright 2021, John Wiley and Sons.

create an integrated cathode/electrolyte structure. Based on this, some additional methods can be employed to further enhance the bonding between cathodes and polymer-based SSEs.

Deng et al. incorporated 20 wt.% of PEGDA-LiTFSI polymer precursor into the LFP cathode slurry and directly cast the slurry onto an aluminum foil, then they cast another layer of PEGDA/PETMP-LiTFSI polymer precursor as the electrolytes and used the UV light to polymerize this integrated cathode/electrolyte structure.^[126] They found that the integrated LFP cathode (mass-loading: 1.76 mg cm⁻²)-PEGDA/PETMP-LiTFSI

SPE (thickness: $\approx 130 \mu\text{m}$) || Li cell demonstrated a discharge capacity of 113.3 mAh g⁻¹ and a capacity retention of 90.2% after 100 cycles under 0.2 C at 60 °C. Yu et al. introduced 10 wt.% methacrylamide-modified gelatin (GelMA)/PEGDA polymer precursor as both the binder and ion conductor into the cathode.^[127] As depicted in Figure 10a, the cathode slurry was cast onto the Al foil and shaped by UV curing for only 30 s, and then the polymer precursor was cast onto the shaped cathode tape and cured by the UV light for another 60 s. The discharge capacity of the integrated LFP cathode-GelMA/PEGDA-LiTFSI GPE ||

Li cell was 136.7 mAh g⁻¹ after 200 cycles at 0.2 C, with a capacity retention of 91.4%, showing obvious improvement than 122.1 mAh g⁻¹ and 80.9% of the conventionally assembled LFP || GelMA/PEGDA-LiTFSI GPE || Li cell. However, UV curing also has light-related limitations. For instance, thick electrolytes may impede light penetration, and the ceramic particles inside the electrolyte can also obstruct the passage of light.

Fusing and hot-pressing can also be used to fabricate the integrated cathode/electrolytes. He et al. prepared the 10% PEO-LiTFSI added LFP cathode and PEO-LiTFSI/LLZO nanowires (PLN) CPE separately. After stacking them together, they fused them at 60 °C to form an integrated structure, as shown in Figure 10b.^[128] The integrated LFP cathode (mass-loading: 1.68 mg cm⁻²)-PLN CPE (thickness: 67 μm) || Li cell achieved a discharge capacity of 158.8 mAh g⁻¹ after 70 cycles under 0.5 C at 60 °C, and 158.7 mAh g⁻¹ after 80 cycles under 0.1 C at 45 °C. He et al. fabricated an integrated cathode-electrolyte structure by adding PVDF-LiTFSI/LATP (PLLA) into the cathode and hot-pressing the cathode and the electrolyte together (Figure 10c).^[129] The integrated LFP cathode (mass-loading: 1.2 mg cm⁻²)-PLLA CPE (thickness: 110 μm) || Li cell achieved a discharge capacity of 149.3 mAh g⁻¹ after 300 cycles under 0.5 C at room temperature with a capacity retention of 93.8%. Even under a current density of 1 C, the cell could still maintain a capacity retention of 91.6% after 400 cycles. It is essential to emphasize the distinction between fusing/hot-pressing and thermal in situ polymerization as they involve different operational targets. Thermal in situ polymerization is directed toward monomers, whereas fusing/hot-pressing is applied to polymers. Additionally, it should be noted that thermosetting polymers are not applicable in this method, as only the thermoplastic polymers can be utilized due to the requirement of polymer re-melting during the fusing and hot-pressing process.

Additionally, electrospinning electrolyte on the cathode presents a novel approach for fabricating integrated cathode-electrolyte structure. In Figure 10d, Zhao et al. proposed an integrated cathode-electrolyte fabrication process by firstly electrospinning PVDF porous network on the cathode tape, and then injecting fiber-reinforced PEO/garnet LLZTO (FPG) CPE into the network.^[130] The electrospinning network ensured the intimate cathode/electrolyte interfacial contact, allowing for the utilization of high-mass-loading cathodes by penetrating the cathode with the FPG CPE solvent. H-F hydrogen bonds were formed between PVDF and PEO, enabling full filling of FPG to the PVDF network as well as reducing the PEO crystallinity. The integrated LFP (mass-loading: 2.5 mg cm⁻²) cathode-PVDF/FPG CPE (thickness: 17 μm) || Li cell had a discharge capacity of 155.2 mAh g⁻¹ under 0.5 C at 45 °C, and the capacity retention was 84.3% after 500 cycles. Even with an increased mass-loading of 13 mg cm⁻², the cell could still achieve a capacity of 124.1 mAh g⁻¹.

4.1.2. Interfacial Electrochemical Stability

Interfacial instability in batteries is another critical challenge for high-performance SSLIBs, which encompasses two key aspects: chemical instability and electrochemical instability. Chemical instability refers to the loss of capability to maintain the original

physical or chemical properties of the interfaces under the condition of no electric field. For example, the polymer materials such as PAN and PVDF demonstrate instability when in contact with lithium metal anodes. Interfacial electrochemical instability refers to the loss of ability to maintain the original properties of the interfaces under the high/low-electric field. To further harness the energy-density in SSLIBs, the batteries are desired to be operated at elevated working voltages. The reduction and oxidation potential of the electrolytes usually determine the lower and upper limit of their electrochemical stability window, and the polymer-based SSEs usually have lower reduction potential and higher oxidation potential compared to the LEs. However, it is still challenging for polymer-based SSEs to meet the high-voltage requirements. If the operating voltage exceeds the upper limit of the electrochemical stability window of the polymer-based SSEs, decomposition of the electrolyte may happen.^[131] Consequently, the electrochemical incompatibility of the cathode/electrolyte interface imposes limitations on the energy-density and cycle performance of the cell. Several strategies have been reported to enhance the electrochemical stability of the electrolyte and enhance the interfacial stability, and we summarize them below.

Building Stable CEI: Building stable CEI between cathodes and polymer-based SSEs is an effective strategy for improving interfacial electrochemical compatibility. Modification for electrolytes, including the incorporation of lithium salt additives, plasticizers, and ceramic fillers can all contribute to the formation of stable CEI.

Lithium salt additives are widely used in building stable CEI as well as preventing interfacial side reactions. For instance, the oxidation of LiDFOB can lead to a stable CEI at the interfacial cathode and enhance the electrochemical compatibility between the high-voltage cathodes and the electrolytes. Other lithium salt additives, such as lithium bis(oxalate)borate (LiBOB) and lithium trifluoro(perfluoro-*tert*-butoxy) borate (LiTFPB) can also contribute to stable CEI formation. Xie et al. designed an ultrathin dual-salt (LiTFSI and LiDFOB) PEO-based SPE.^[132] TEGDME was introduced to generate cross-linked networks with PEO by in situ UV curing. The preferential decomposition of LiDFOB led to the formation of a stable Li_xBO_yF_z and LiF-rich CEI layer on the surface of LiNi_{0.6}Co_{0.2}Mn_{0.2}O₂ (NCM622) cathodes. Notably, an ultrathin in situ polymerized PEO-based SPE, as well as a good cathode wetting, were achieved. The NCM622 (mass-loading: ≈1.5 mg cm⁻²) || dual-salt PEO-based SPE (thickness: 20 μm) || Li cell delivered an initial discharge capacity of 169.2 mAh g⁻¹ and capacity retention of 94.7% after 100 cycles under 0.1 C at 30 °C with the cut-off voltage of 4.2 V. Li et al. proposed a ring-opening polymerized PDOL-based dual-salt (LiTFSI and LiDFOB) GPE, among which LiDFOB could act as initiator.^[133] The two salts both contributed to forming stable inorganic-rich (F-, N-, and B-species) CEI and SEI, as shown in Figure 11a. The LiMn₂O₄ (mass-loading: 1.5 mg cm⁻²) || dual-salt PDOL-based GPE || pretreated-Li had an initial discharge capacity of 98.7 mAh g⁻¹ and a capacity retention of 92.5% after 126 cycles under 0.1 C at room temperature with the cut-off voltage of 4.3 V.

In addition to lithium salt additives, the incorporation of plasticizers and ceramic fillers can also contribute to the formation of stable CEI. Zhang et al. designed a polymer-in-“quasi-ionic

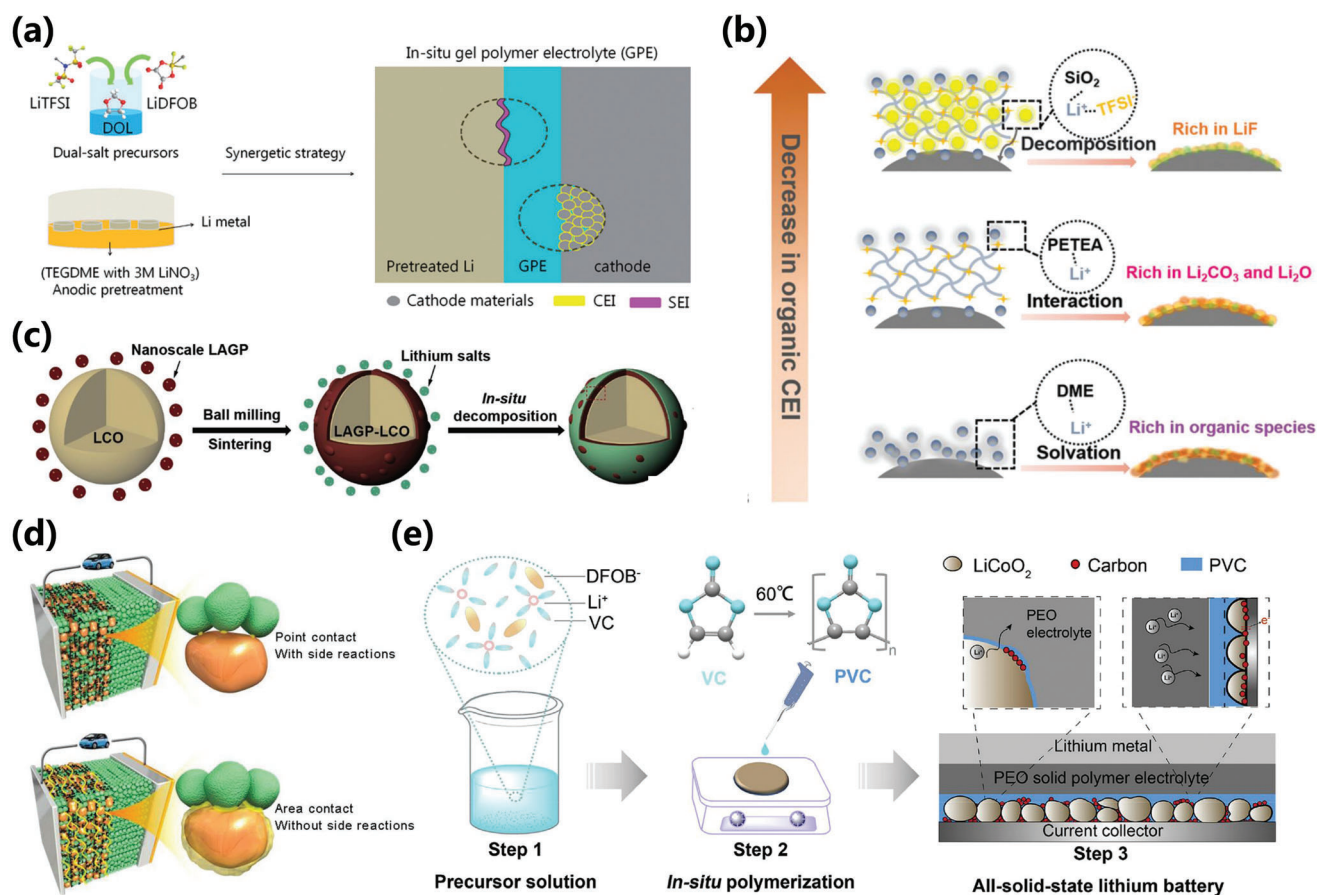


Figure 11. a) Building stable CEI and SEI by incorporating LiTFSI/LiDFOB dual-salt to the in situ polymerized PDOL GPE. Copyright 2020, Royal Society of Chemistry. b) Building stable LiF-rich CEI by adding SiO₂ fillers to PETEA-LiTFSI/DME/DOL GPE. Reproduced with permission.^[16] Copyright 2023, John Wiley and Sons. c) Enhancing the LCO/PEO interfacial stability via the synergetic LAGP nano-layer coating and the in situ CEI formation. Reproduced with permission.^[136] Copyright 2020, Elsevier. d) Improving the interfacial contact and suppressing the side reactions by coating NCM622 with the PAB polymer layer. Reproduced with permission.^[137] Copyright 2018, John Wiley and Sons. e) Improving the LCO/PEO interfacial stability via the PVC-LiDFOB coating. Reproduced with permission.^[138] Copyright 2020, Elsevier.

liquid" electrolyte (PQILE) by mixing PEO with the lithium DME-bis(fluorosulfonyl) imide (LiFSI) quasi-ionic liquid (so-called solvated ILs, in which ligand molecules are coordinated with salt cations/anions to form complex ions).^[134] The addition of DME reduced the electron density of the ether oxygen in PEO and the ether solvent molecules could help form a stable LiF-rich CEI on the surface of LiNi_{0.33}Co_{0.33}Mn_{0.33}O₂ (NCM111) cathode. The resultant NCM111 (areal loading: 2.0 mAh cm⁻²) || PQILE || Li cell achieved the capacity retention of 88.4%, 86.7%, and 79.2% after 300 cycles at the charging cut-off voltages of 4.2, 4.3 and 4.4 V, respectively, under a charging current of C/3 and a discharge current of 1 C at 60 °C. As mentioned in Section 2.2.4, the incorporation of SiO₂ fillers into PETEA-LiTFSI/DME/DOL GPE facilitated the formation of AGGs-dominated solvation sheaths, and therefore promoted the formation of stable inorganic LiF-rich CEI and SEI, as depicted in Figure 11b.^[16] This phenomenon can be explained based on the discussion in Section 2.1.1, which suggests a transition from solvent molecules-dominated solvation sheaths to lithium salt anions-dominated solvation sheaths. As a result, the LFP (mass-loading: 5 mg cm⁻²) || PETEA-LiTFSI/DME/DOL/SiO₂ CGPE || Li cell displayed an initial ca-

capacity of 147 mAh g⁻¹ and even after 700 cycles under 0.5 C at 50 °C, the cell still had a capacity retention of 68.5%.

Coating on Cathode: As discussed in Section 2.1.3, ICEs offer outstanding electrochemical stability, thus the coating of ICEs on the cathodes can be used as man-made stable CEIs between the cathodes and the polymer-based SSEs within the kind of sandwich structure. Furthermore, certain polymers with superior electrochemical stability can also serve as effective coatings.

Li et al. coated LCO with an ultrathin (<20 nm) LATP layer.^[135] With the cut-off voltage of 4.2 V, the LCO-LATP || PEO-LiTFSI SPE || Li cell exhibited an initial discharge capacity of 176 mAh g⁻¹ with a capacity retention of 93.2% after 50 cycles under 0.2 C at 60 °C, suggesting that the LATP coating layer is effective in suppressing the oxidation of PEO at high-voltage. Similarly, Yang et al. coated the LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ (NCM523) cathode with a LAGP layer by scalable ball milling and sintering (Figure 11c).^[136] In addition to LiTFSI, LiBOB, and LiPF₆ were also added to the PEO/PEGDME electrolyte to further increase the interfacial electrochemical stability by their decomposition in the pinholes of LAGP coatings. Employing a cut-off voltage

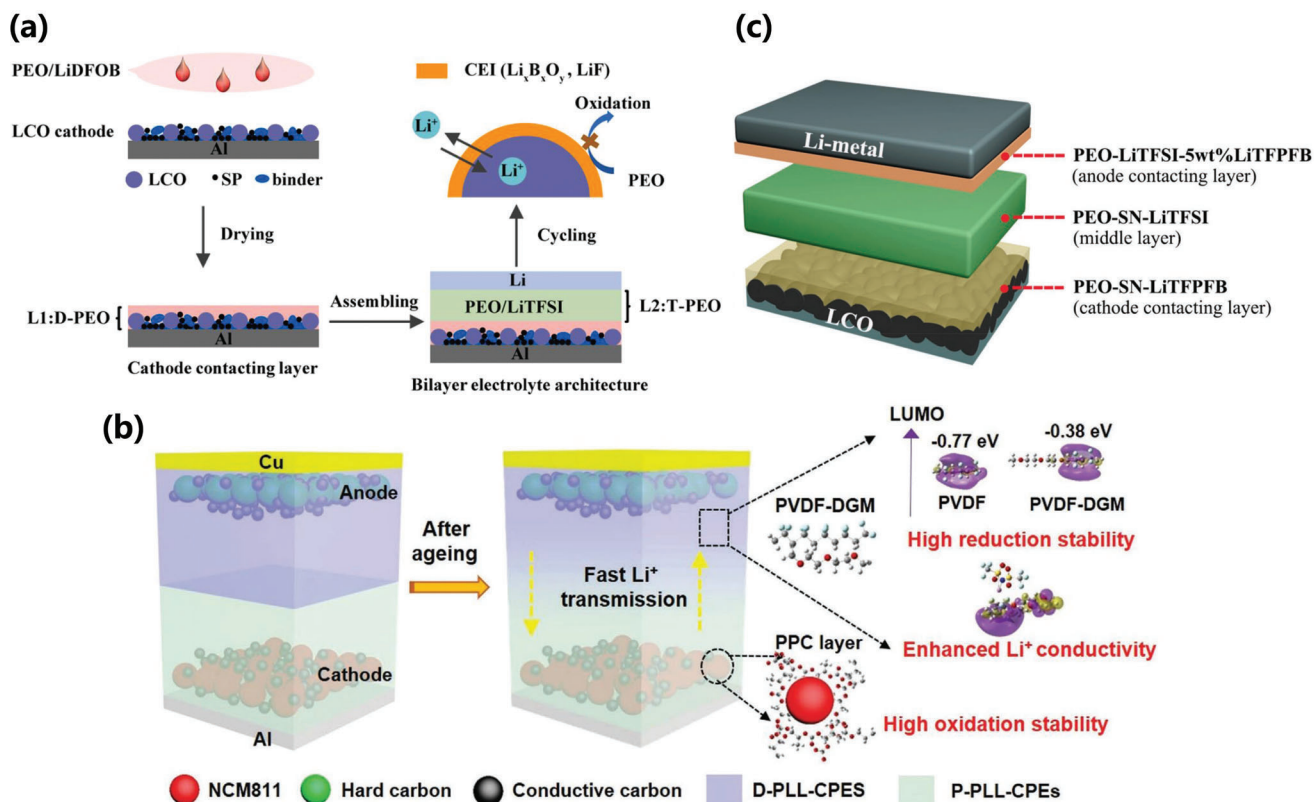


Figure 12. a) Synthesis of bilayer electrolytes with different lithium salts and stable CEI formation at the cathode/electrolyte interface. Reproduced with permission.^[139] Copyright 2023, Elsevier. b) Fabrication process of quasi-double-layer electrolytes with different plasticizers and their roles. Reproduced with permission.^[140] Copyright 2022, John Wiley and Sons. c) Structure of multilayer PEO-based SPE. Reproduced with permission.^[142] Copyright 2019, John Wiley and Sons.

of 4.25 V, the NCM523-LAGP (mass-loading: $\approx 10.5 \text{ mg cm}^{-2}$) || PEO/PEGDME-LiTFSI-LiBOB-LiPF₆/EC/PC GPE || Li cell displayed an initial discharge capacity of 131.2 mAh g^{-1} and capacity retention of 81.9% after 400 cycles under 0.5 C at 60 °C.

The polymer coatings can not only contribute to enhanced interfacial electrochemical stability but also establish a close cathode/electrolyte interfacial contact. Guo et al. coated a poly(acrylonitrile-co-butadiene) (PAB) layer on the NCM622 cathode.^[137] As shown in Figure 11d, the polymer layer improved the cathode/electrolyte interfacial contact and also prevented the interfacial side reactions. In addition, the -CN group of PAB favored the Li⁺ conduction at the interface. With a cut-off voltage of 4.3 V, the NCM622-PAB (mass-loading: 2 mg cm^{-2}) || poly(ether-acrylate)-LiPF₆ GPE || Li cell had a discharge capacity of 99 mAh g^{-1} under 3 C at 60 °C, and the capacity retention was 75% after 400 cycles under 1 C at 60 °C. Li et al. introduced a 5-nm poly(vinylene carbonate) (PVC)-LiDFOB coating on the LCO cathode with the process shown in Figure 11e.^[138] With a discharge cut-off voltage of 4.2 V, the LCO-PVC-LiDFOB (mass-loading: $\approx 2.5 \text{ mg cm}^{-2}$) || PEO-LiTFSI SPE || Li cell achieved a discharge capacity of 142.0 mAh g^{-1} under 0.5 C at 60 °C, with a capacity retention of 71.5% after 500 cycles.

Symmetric/Asymmetric Multilayer Electrolyte: Multilayered polymer-based SSEs offer a promising approach to protect electrochemically incompatible polymers from direct contact with cathodes. Wang et al. designed a bilayer PEO-based electrolyte by

introducing an ultrathin ($\approx 1 \mu\text{m}$) PEO-LiDFOB layer between the LCO cathode and the PEO-LiTFSI electrolyte (Figure 12a).^[139] The PEO-LiDFOB layer facilitated the formation of a stable CEI containing Li_xB_yO_z and LiF, effectively suppressing the oxidation of PEO under high voltages. Furthermore, an intimate cathode/electrolyte contact was constructed in situ by casting the PEO-LiDFOB precursor directly onto the LCO surface. On the anode side, the PEO-LiTFSI layer provided high-ionic conductivity and stabilized the electrolyte/lithium metal anode interface. Under a cut-off voltage of 4.2 V, the LCO (mass loading: 1.5 mg cm^{-2}) || bilayer electrolyte || Li cell exhibited an initial discharge capacity of 125 mAh g^{-1} and a capacity retention of 75% after 100 cycles under 0.2 C. Huang et al. prepared a quasi-double-layer PVDF/PVDF-HFP-LiTFSI/LLZO CGPE containing two different plasticizers: the oxidative stable PC and the reductive stable diethylene glycol dimethyl ether (DGM).^[140] Specifically, the DGM-added layer was toward the anode side, while the PC-introduced layer was toward the cathode side. As depicted in Figure 12b, the interface between the two electrolyte layers could be erased after ageing. The addition of PC enabled a conformal coating on the contours of the LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NCM811) particles, just like an artificial CEI layer with a thickness of 15–30 nm, thereby enhancing the antioxidant ability. The NCM811 (mass-loading: $2.0\text{--}3.0 \text{ mg cm}^{-2}$) || quasi-double-layer electrolyte || hard carbon cell delivered a discharge capacity of 139.2 mAh g^{-1} under 0.1 C at room temperature after 50 cycles.

Even when the current density was increased to 1 C, its capacity retention remained at 80.2% after 200 cycles at room temperature. Wang et al. constructed a symmetric polymer electrolyte by introducing the Pyr₁₃TFSI IL-containing PVDF-HFP-based electrolyte (IPL) on both sides of the PEO-LiTFSI/LLZTO, with a total thickness of $\approx 70\text{ }\mu\text{m}$.^[141] The IPL layers could suppress the interfacial side reactions and facilitate the formation of a stable CEI under high voltage. The initial discharge capacities of the LCO (mass-loading: 1.5 mg cm^{-2}) || symmetric IPL-PEO-LiTFSI/LLZTO-IPL electrolyte || Li cells were 144.3 mAh g^{-1} and 169.7 mAh g^{-1} under 0.2 C with the cut-off voltages of 4.3 V and 4.5 V, respectively, and the capacity retentions after 100 cycles were 85.9% and 82.7%. Cui et al. exploited an asymmetric multilayer PEO-based SPE by introducing cathode and anode contacting layers containing LiTFPB (Figure 12c).^[142] The LiTFPB contributed to the formation of stable CEI and SEI and also facilitated the complexation between Li⁺ and ether oxygen. The ether oxygen could donate the lone electron pairs to Li⁺, thus enhancing the oxidation potential of PEO and further improving the cathode/electrolyte interface electrochemical compatibility. As a result, with the cut-off voltage of 4.3 V, the LCO (mass-loading: 5 mg cm^{-2}) || asymmetric multilayer PEO-based SPE || Li cell achieved an initial discharge capacity of 150 mAh g^{-1} and capacity retention of 83.3% after 100 cycles under 0.1 C at 60 °C. Although the multilayer design offers potential benefits in terms of improved interfacial stability, careful attention must be paid to controlling the thickness of the electrolyte to mitigate negative impacts on the energy-density of SSLIBs.

4.1.3. Interfacial Space-Charge Layer

Large interfacial resistances may still exist due to the formation of space-charge layers (SCLs) even if the cathode/electrolyte interface has intimate contact and good electrochemical compatibility. This is because when two materials with different chemical potentials come into contact, such as the electrodes and polymer electrolytes, space charges will accumulate at the interface. In addition to the different chemical potentials, the electric potential difference between the electrodes and the polymer electrolytes drives the mobile charge to redistribute, leaving Li⁺ accumulating/lacking layers at the interfaces. The SCLs at the interface may increase the energy barriers for Li⁺ conduction, resulting in a high resistance at the interface,^[143] and they will grow with the increased electric potential difference between the cathodes and the electrolytes.^[144]

The introduction of the ferroelectric/dielectric materials, such as inorganic K_{0.5}Na_{0.5}NbO₃ (KNN), LiNbO₃ (LNO), BTO, SrTiO₃ (STO), and organic-inorganic hybrid guanidinium perchlorate (GClO₄) into the interfaces can be an excellent solution to suppress the interfacial accumulated space charges. Wang et al. introduced a thin layer of ferroelectric KNN at the cathode/electrolyte and the anode/electrolyte interfaces.^[145] Upon charging, the polarization of the KNN layers was switched, leading to the establishment of a local electric field. This electric field could mitigate the interfacial potential gap and suppress the SCLs, facilitating interfacial Na⁺ conduction, as illustrated in Figure 13a. The interfacial ferroelectric-engineered Na₃V_{1.85}Fe_{0.15}(PO₄)₃ (NVP, mass-loading: $0.9\text{--}1.1\text{ mg cm}^{-2}$) || PEO/NZSP CPE || Na cell demon-

strated a high-discharge capacity of 160.3 mAh g^{-1} and a retention of 97.4% after 165 cycles at room temperature. In addition, the ferroelectric-engineered battery had been suspended for 2 months during the charge/discharge cycle, and when it resumed the cycle, a high-capacity retention of 86% could still be achieved after 180 cycles. Tateyama et al. incorporated a ferroelectric LNO buffer layer between the LCO cathode and the β -Li₃PS₄ electrolyte.^[146] The inclusion of the ferroelectric buffer layer resulted in the formation of a smooth interface and effectively reduced the interfacial SCLs. During battery charging, the ferroelectric buffer layer successfully impeded the growth of the SCLs and facilitated efficient Li⁺ conduction. Cui et al. coated a discontinuous BTO at the LCO cathode to inhibit the formation of the SCLs and promote Li⁺ conduction at the interface with the LPSCl electrolyte.^[144] Along with the polarization orientation, the negative side of the BTO dipoles faced the positive charges on the cathode side while the positive side of BTO faced the opposite direction. In this way, the interfacial Li⁺ was redistributed by the BTO-induced electric field, suppressing the space-charges accumulation, as depicted in Figure 13b. The total interfacial resistance of the LCO-BTO (mass-loading: 8.9 mg cm^{-2}) || LPSCl || In-Li cell was reduced from $2993\text{ }\Omega$ to $1339\text{ }\Omega$. Additionally, the cell exhibited a high-discharge capacity of nearly 140 mAh g^{-1} under 0.2 C at 30 °C. Even under a high-current density of 1 C, it still displayed a discharge capacity of 92 mAh g^{-1} , much $> 60\text{ mAh g}^{-1}$ of the LCO || LPSCl || In-Li cell.

Besides the inorganic BTO and STO, the organic-inorganic hybrid ferroelectric guanidinium perchlorate (GClO₄) can also be used at the interfaces to prevent SCLs.^[147] GClO₄ coating has a single-domain state with an upward self-polarization effect, creating a built-in electric field perpendicular to the cathode/electrolyte interface to alleviate the SCLs, as depicted in Figure 13c. The self-polarization was primarily attributed to the flexoelectric effect arising from the lattice mismatch between the GClO₄ coating and the LCO cathode. The LCO-GClO₄ (mass-loading: 8.9 mg cm^{-2}) || LPSCl || In-Li cell showed a discharge capacity of 210.6 mAh g^{-1} under 0.1 C at 30 °C, and the capacity retention was 63.2% after 100 cycles.

To address the SCLs issue at the anode, Li et al. proposed a Kevlar-derived hydrogel (KevlarH) electrolyte containing aramid nanofibers and ZnSO₄.^[148] The amide group on the aramid chain in KevlarH was able to immobilize SO₄²⁻ and reduce the concentration polarization of Zn²⁺ at the interface by preventing the depletion of SO₄²⁻ (Figure 13d), thus reducing the SCLs effects at the anode/electrolyte interface. There was also a relatively weak interaction between the Zn²⁺ and the carbonyl group of the aramid chain, which redistributed the Zn²⁺ flux and enabled the uniform Zn deposition without reducing their diffusion rate. These characteristics of the KevlarH electrolyte offered a promising approach to address the lithium dendrite formation and improve the stability of zinc-based batteries.

The aforementioned strategies all demonstrate their potential in improving the cathode/electrolyte interfaces, thereby contributing to the achievement of high-performance SSLIBs. However, it is worth noting that the cathode mass-loading in the reported cells is $1\text{--}5\text{ mg cm}^{-2}$, far below the requirement for high-energy-density SSLIBs. Although a few cases have

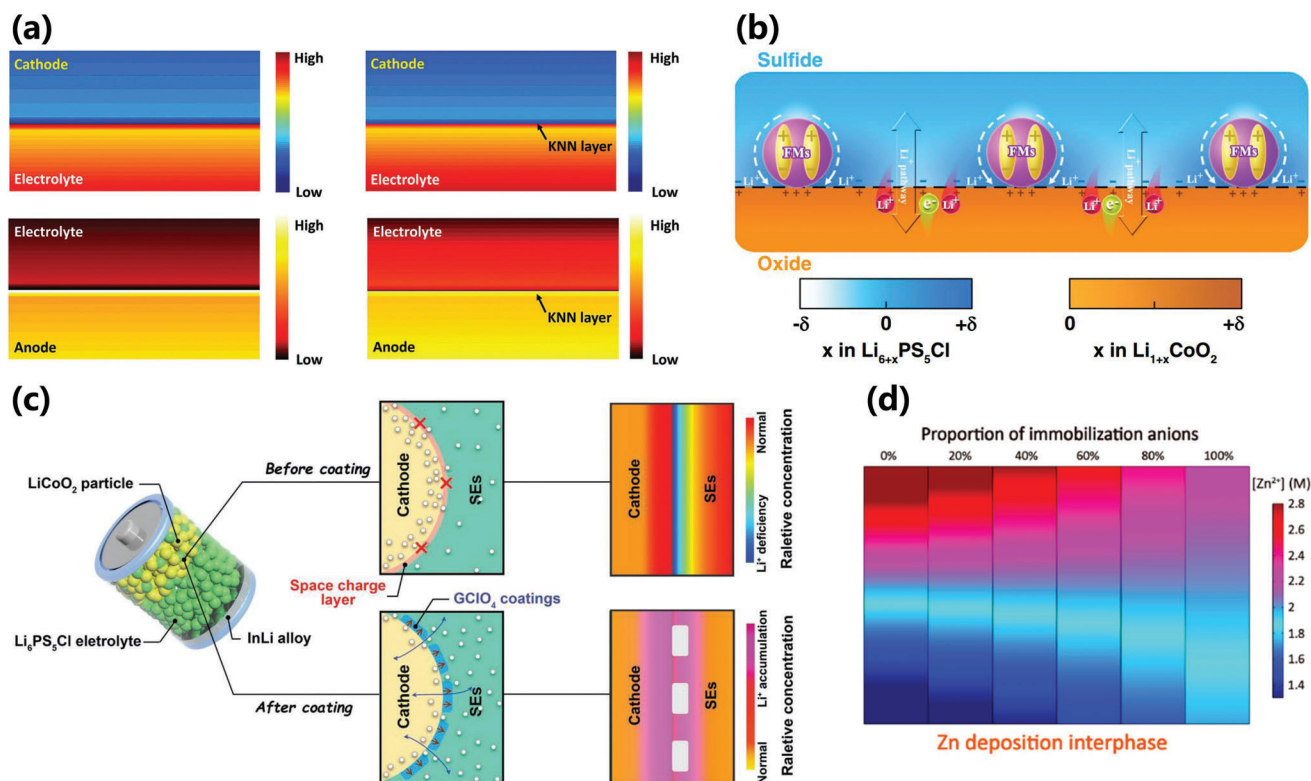


Figure 13. a) Simulated Na⁺ concentration at the electrode/electrolyte interfaces with/without the KNN layer. Reproduced with permission.^[145] Copyright 2022, John Wiley and Sons. b) Schematic illustration of SCLs with BTO-engineered interface. Reproduced with permission.^[144] Copyright 2020, Springer Nature. c) Schematic illustrations of interfaces and finite element simulation of charge concentrations before and after GClO₄ coating. Reproduced with permission.^[147] Copyright 2023, John Wiley and Sons. d) Zn²⁺ concentration distribution at the anode/electrolyte interface with different amounts of immobilized SO₄²⁻. Reproduced with permission.^[148] Copyright 2023, American Chemical Society.

suggested the possibility of achieving higher mass-loading (>10 mg cm⁻²), it often comes at the expense of discharge capacity and cycle performance.^[124,130,149,150] Recent advancements have seen the emergence of extremely high-mass-loadings, reaching 50 mg cm⁻² or even 100 mg cm⁻², while maintaining acceptable cycle performance.^[151,152] However, these designs are still based on LEs. If similar cathode mass-loading can be achieved in SSLIBs, it can definitely pave the way for a new era.

4.2. Anode/Polymer-Based SSE Interfaces

The anode materials and the anode/electrolyte interfaces also determine the energy-density of SSLIBs. Among various anode materials, lithium metal exhibits a significantly higher specific capacity (3860 mAh g⁻¹) compared to graphite (372 mAh g⁻¹). Moreover, lithium metal possesses a low reduction potential (3.04 V vs standard hydrogen electrode).^[153,154] These characteristics can enhance the energy-density of lithium metal anode SSLIBs by 40% to 50% compared to the ones using graphite anodes.^[155] Silicon anodes also demonstrate high specific capacity, however, they suffer from substantial volume expansion during cycle, thereby compromising the stability of the SEI and negatively impacting the battery performance.^[156–158] In recent years, anode-free SSLIBs have emerged as a potential av-

enue for future development, as they demonstrate even higher energy-density. Nevertheless, its practical implementation is hindered by the issues such as dendritic lithium plating on current collectors and low coulombic efficiency.^[159–161] Therefore, at the current stage, lithium metal is still the “holy grail” of high-energy-density SSLIB anodes. The lithium metal anode/electrolyte interfaces also present several challenges. These challenges not only impact the energy-density of SSLIBs but also affect their cycle performance and safety. Consequently, it is necessary to optimize the lithium metal anode/electrolyte interfaces to address these issues.

4.2.1. Interfacial Chemical Stability

Interfacial stability is a prerequisite for a cyclic stable SSLIB. However, the lithium metal anodes will always react with the polymer-based SSEs containing specific groups or elements, such as the —C≡N group of PAN, titanium from the LATP and LLTO fillers, and germanium from the LAGP fillers, as mentioned in the previous section.^[33–35,95] Fortunately, the robust SEI that can only transport Li⁺ can suppress the side reactions and improve the interfacial chemical stability. For example, the addition of LiDFOB or FEC can induce the formation of LiF-rich SEI and protect the polymer-based SSEs from

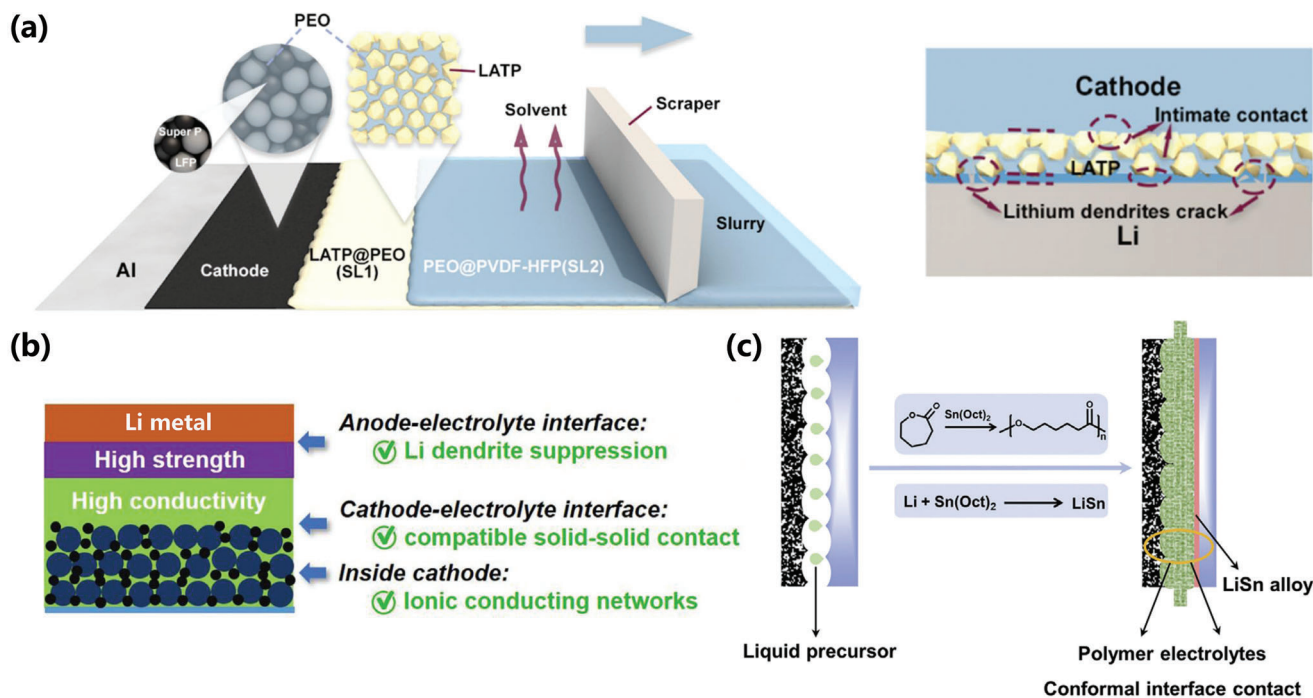


Figure 14. a) PEO/LATP and PEO/PVDF-HFP bilayer electrolyte design to improve the lithium metal anode/electrolyte interfacial stability. Reproduced with permission.^[150] Copyright 2021, Elsevier. b) PEO/10 wt.% LLZTO and PEO/40 wt.% LLZTO bilayer electrolyte design to suppress the lithium dendrites. Reproduced with permission.^[162] Copyright 2020, American Chemical Society. c) In situ ring-opening polymerization and formation of LiSn layer to suppress the lithium dendrites. Reproduced with permission.^[163] Copyright 2021, Elsevier.

the reduction of the lithium metal anodes.^[76,133] Moreover, the multilayer design, such as the Janus structure mentioned in Section 3.2, could physically separate the reactive components from the lithium metal anodes.^[98] He et al. fabricated an integrated structure of cathode/bilayer polymer electrolytes (Figure 14a).^[162] They designed a thin layer of PEO/PVDF-HFP-LiTFSI toward the lithium metal anode, thus hindering the side reactions with LATP. The integrated LFP cathode-PEO-LiTFSI/LATP and PEO/PVDF-HFP-LiTFSI bilayer electrolyte || Li cell demonstrated an outstanding capacity retention of 91.7% after 100 cycles.

4.2.2. Lithium Dendrite Suppression

Most importantly, the uneven deposition of Li^+ at the interface gives rise to the formation of lithium dendrites, which can subsequently pierce the electrolyte and result in short circuits of the LIBs. This poses a significant threat to the safety and cycle performance of the SSLIBs using polymer-based SSEs. In severe cases, this issue can even trigger a thermal runaway if the electrolyte contains flammable components. This concern is particularly pronounced in high-energy-density SSLIBs, where ultra-thin polymer-based SSEs are preferred, thereby amplifying the risks. Compared to SPEs and GPEs, CPEs and CGPEs exhibit enhanced mechanical strength by incorporating ceramic fillers. Especially, the all-solid-state CPEs demonstrate superior capability in suppressing the growth of lithium dendrites. It is generally accepted that increasing the content of ceramic fillers can fur-

ther strengthen the electrolyte and improve its ability to suppress the lithium dendrites. However, as discussed in Section 2.2.3, the “polymer-in-ceramic” CPE structure with a high ceramic content may suffer from the severe agglomeration of the ceramic fillers, thereby compromising its ionic conductivity. To solve this problem, Guo et al. devised a bilayer electrolyte design (PEO-LiTFSI/10 wt.% LLZTO and PEO-LiTFSI/40 wt.% LLZTO).^[150] As illustrated in Figure 14b, the PEO-LiTFSI/40 wt.% LLZTO layer with superior mechanical strength is positioned toward the lithium metal anode. Under 0.2 mA cm^{-2} at 30°C , the Li || bilayer electrolyte || Li cell could cycle stably for 400 h, while the cell with single-layer PEO-LiTFSI/10 wt.% LLZTO electrolyte occurred short circuit after only 147 h cycle. In addition, changing the morphology of ceramic fillers, such as using nanowires or even building a ceramic network inside the CPEs can further enhance their lithium dendrite suppression ability. For instance, using the CPE mentioned in Section 3.1 (Figure 7a), the Li || PEO-LiTFSI/LLZO/PVA nanofiber network CPE || Li cell could cycle stably for 1000 h under 0.5 mA cm^{-2} at 25°C .^[74] On the other hand, the in situ formed interphase with enhanced mechanical strength could also effectively suppress the lithium dendrites. For example, Xie et al. reported an in situ polymerization method based on ϵ -caprolactone (ϵ -CL) ring-opening polymerization.^[163] They selected an initiator of $\text{Sn}(\text{Oct})_2$ which could induce the polymerization and also interact with the lithium metal to form a LiSn interphase layer at the anode interface and suppress lithium dendrites. The pre-assembled Li || PCL-LiTFSI/PC GPE || Li cell cycled stably for 900 h under 0.05 mAh cm^{-2} at 60°C with only a slight increment of overpotential. In comparison, the cell with

ex-situ polymerized electrolyte underwent a short circuit only after a 200 h cycle.

5. Conclusions and Perspectives

The growing demand for high-performance portable electronic devices and electric vehicles has underscored the importance of high-energy-density LIBs. Among the various electrolytes available, polymer-based SSEs offer a comprehensive performance that aligns with the requirements of high-energy-density LIBs. This can be attributed to their thinner and lighter structure, as well as their ability to establish superior electrode/electrolyte interfacial contact. To further enhance the ionic conductivity and thermal/electrochemical/mechanical stability of polymer-based SSEs, additives including LEs, ILs, and ICEs have been introduced, leading to the development of GPEs, CPEs, and CGPEs. This review provides a summary of the comprehensive performance of these electrolytes, focusing particularly on the Li^+ conduction within each electrolyte. Such understanding can aid in the design of electrolytes with higher theoretical ionic conductivity. Additionally, the review presents a comparison of the advantages and disadvantages associated with the most commonly used polymer matrix materials, including PEO, PAN, PVDF, and PMMA. Furthermore, typical modification strategies aimed at improving the shortcomings of these polymer matrix materials are introduced. Achieving a good cathode/electrolyte interface is a critical factor in attaining high-energy-density SSLIBs. This review analyzes the significant issues that impact the interface between cathodes and polymer-based SSEs, including interfacial contact and electrode wetting, interfacial electrochemical stability as well as interfacial space-charge layers. To enhance the cathode/electrolyte interfacial contact and electrode wetting, two promising approaches are highlighted: in situ polymerization and integrated cathode/electrolyte structures. These methods enable the polymer electrolytes to effectively penetrate the thick cathodes and enhance the interfacial contact. Addressing the electrochemical compatibility between polymers and the cathodes can be accomplished through building stable CEIs, introducing coatings on cathodes, or constructing symmetric/asymmetric multilayer electrolyte structures. Additionally, for the emerging interfacial issue of SCLs, the application of interfacial ferroelectric/dielectric engineering is a promising solution. The anode materials and lithium metal anode/electrolyte interfaces also significantly influence the energy-density of SSLIBs. This review summarizes the anode/electrolyte interfacial issues including the interfacial chemical stability and lithium dendrite suppression. Building stable SEI can handle the chemical instability, and regulating the ceramic content of electrolytes can help suppress the lithium dendrites. Designing suitable multilayer electrolytes can effectively address the two problems. The electrode/electrolyte interface issues in SSLIBs with polymer-based SSEs and their remedies are all summarized in **Figure 15**.

Although polymer-based SSEs have demonstrated great potential for the next generation of high-energy-density SSLIBs, several challenges still need to be well addressed to further enhance the battery performance:

- 1) The transport mechanism of Li^+ in polymer-based SSEs remains incompletely understood, particularly for SSEs con-

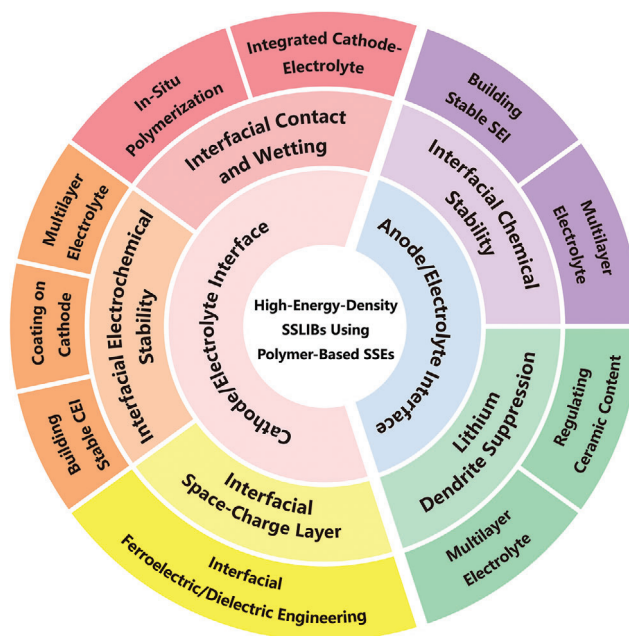


Figure 15. Electrode/electrolyte interface issues and remedies in SSLIBs with polymer-based SSEs.

sisting of multiple components including GPEs, CPEs, and CGPEs. This lack of understanding hinders the further enhancement of ionic conductivity. Among these SSEs, all-solid-state CPEs are considered as the future development direction. However, due to the absence of liquid phase additives, the ionic conductivity of CPEs is generally lower compared to GPEs and CGPEs, making it insufficient for meeting the practical application demands of SSLIBs. Consequently, deep investigation into the Li^+ transport mechanism in polymer-based SSEs is necessary to facilitate the novel designs of electrolytes with superior ionic conductivity.

- 2) The comprehensive properties of the polymer-based SSEs still require further improvement, particularly for the high-energy-density SSLIBs. The wide electrochemical stability window of polymer-based SSEs plays a crucial role in ensuring their compatibility with high-voltage and high-capacity cathodes. Additionally, the thickness/mass of the electrolytes significantly impacts the energy-density of SSLIBs. However, thinner electrolytes possess lower mechanical strength, rendering them susceptible to puncture by lithium dendrites. This poses a safety challenge for SSLIBs. Hence, future research efforts should focus on improving the electrochemical stability window and mechanical strength of polymer-based SSEs. Additionally, exploration of new materials exhibiting superior performance or novel structure design of polymer-based SSEs is still necessary to further enhance the energy-density, cycle performance, and safety of SSLIBs.
- 3) The mass-loadings of cathodes have a direct impact on the energy-density of SSLIBs. However, the so-far reported works are still below satisfactory. In future research, it is crucial to improve the mass-loadings of the cathodes while ensuring the discharge capacity and cycle performance of batteries are not compromised. Moreover, aiming at further reducing

the battery mass, anode-free SSLIB designs should receive more attention. On the other hand, the cathode/electrolyte and anode/electrolyte interfacial challenges, including interfacial contact, stability, resistance, and strength, greatly influence the performance of high-energy-density SSLIBs. Although there are individual solutions for each issue, no reported strategy has successfully addressed all the problems simultaneously. In the future, the interfacial side reaction mechanism, ion conduction mechanism, and dendrite formation/suppression mechanism should be well explored to further improve the battery performance.

- 4) Despite the achievements received in the polymer-based SSEs, their production remains limited in the laboratory and needs high cost. Therefore, it is imperative for future studies to prioritize the development of cost-effective polymers that can be easily synthesized. Efforts should also be made to streamline the fabrication process of SSLIBs using such polymer-based SSEs.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

electrode/electrolyte interfaces, high-energy-density, polymer-based solid-state electrolytes, solid-state batteries

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